Resolution Geometry

Indistinguishability, Its Symmetries, and the Geometry of What Observations Determine

Matěj Rada
MatejRada@email.cz
Licensed under CC BY-NC-ND 4.0

October 2026

Abstract

We develop Resolution Geometry (RG), a framework for the geometry of what a declared family of observation protocols can and cannot determine about a latent system. The primitive is the equality-of-law relation of the joint observation law. This relation is a groupoid whose orbit space is the observable quotient. Modulo its gauge part, the stabilizer of the law is an effective symmetry group acting faithfully on the quotient.

Every further structure has one of three origins:

- it is constructed canonically from these objects once its law-space definition is fixed;
- it is characterized by an established theorem under a stated selection principle;
- it is conditioned on a named and testable hypothesis.

All statements are relative to the specified observation system, and the exact layer is conditional on the specified law map. The further structures are, in order:

- differential and stratified structure;
- statistical (Fisher, Hellinger, Le Cam), quantum (Petz, Gleason) and probabilistic structure;
- causal and characteristic structure;
- scale and transport;
- duration and dynamics;
- calibrated physical distance.

The theory is organized as a ladder of rungs with backward-only dependencies, and is assembled by a kernel–stabilizer theorem.

Across walls where distinguishability is lost, the structure continues by non-invertible specialization. The global transport problem splits into a closed observable atlas and a latent connection that is canonical only under declared second-order data. Uncertainty is shown to be a consequence of indistinguishability rather than a primitive. Probability is forced only by exchangeability, mixing on a declared regularity class, or quantum structure. Priors are forgotten on the quotient and never along the observational kernel. Gaussian propagation is the LAN limit on regular strata. Monte Carlo is objective exactly on outputs certified invariant over a declared admissible class.

Chaotic complexity is not absorbed by declarations. The boundary between certified and uncertified forecasts has slope $1/\lambda$ in the logarithm of declared precision. Information about the past grows at rate 2λ, and observable quotients grow at rate h. Protocols move intercepts only, and the precision cost per certified step is $\lambda/\ln 2$ bits (Brudno, Pesin).

Every admitted request ends in exactly one typed status, with false-lock probability bounded by the gate errors. We also give the computational complexity of each layer. The remaining open problems are one problem of real algebraic geometry and uniform forgetting for forced deterministic turbulence. Every RG-specific statement has a deterministic witness and a mutation canary in an accompanying test suite.

Contents

1 Introduction **2**

2 Conventions **2**

3 Objects **4**

4 The exact layer **4**

5 The symmetry layer **5**
5.1 The observable sector and source-side automation 6

6 The differential layer **7**
6.1 Specialization across walls 8

7 The second-order layer **9**
7.1 The quantum metric module 10

8 Ignorance and the observation limit **11**

9 Uncertainty, probability and forgetting **12**

10 Statistical distance and angle **13**

11 Order, cone and conformal class **14**
11.1 The residual characteristic branch 15

12 Scale and transport **15**
12.1 Canonical transport 16

13 Duration **17**
13.1 Dynamics 17
13.2 Chaos is not hidden: the frontier law 18

14 Physical distance **19**
14.1 No internal unit 19
14.2 Displacement and invariant metric 19
14.3 Anchor and identification 20

15 The assembled theorem **21**

16 Certificates and closure **22**

17 Computational complexity **24**

18 The ladder **24**

19 Procedure **28**

20 Status **29**

21 Open problems **29**

1 Introduction

Every measurement program faces two questions before any estimate is computed. What does the declared experiment determine? And which further structures (metric, causal order, units, transport, probability) does it force, rather than presuppose? Information geometry [1, 5] answers part of the second question for regular parametric families, and comparison of experiments [3, 11] answers part of the first. Identifiability theory, measurement theory [10], the order-theoretic reconstruction of spacetime [8, 13] and ergodic theory [43, 40] each treat one layer. They are rarely combined, and the hypotheses that connect them are usually implicit.

RG makes those connections explicit. Its single primitive is the equality-of-law relation of a declared joint observation law. Everything else is placed on a *ladder*: a dependency-ordered list of rungs, each *forced* by earlier rungs, *imported* from an established theorem, conditioned on a named *hypothesis*, conditional on an effective symmetry group, a *no-go*, or *open*.

Main results.

1. *Kernel–stabilizer assembly* (Theorem 15.1). Once its law-space definition is fixed, every law-defined structure descends canonically to the quotient and is invariant under the effective symmetry. Selecting that definition is a separate step, made either by a stated characterization (Markov invariance for Fisher) or by a declaration.

2. *Stratified closure* (Sections 6.1, 16). Walls are crossed by non-invertible specialization. Every admitted request ends in exactly one typed status, sound up to the gate errors.

3. *Observable sector* (Section 5.1). The complete antichain of minimal sufficient protocol panels can be selected without hidden preference. Adding protocols intersects kernels; requiring observability under more models closes their union.

4. *Uncertainty* (Section 9). Uncertainty splits along the quotient. Probability, Gaussian propagation and Monte Carlo are theorems with stated hypotheses, and non-circularity is a certificate over a declared class.

5. *Chaos* (Section 13.2). Chaotic exponents appear as slopes of RG's own decision boundaries, which no declaration can change, and they carry an explicit information cost.

6. *Physical distance* (Section 14). The metre is reached only through rigid counted channels, anchors and declared media. Stationary slices carry the Landau–Lifshitz radar metric, and non-stationary slices carry none.

Section 17 gives the computational complexity of each layer. Section 18 lists the rungs, and Section 21 the open problems.

2 Conventions

Scope. Unless stated otherwise, latent spaces are finite-dimensional smooth manifolds, experiments are classical, dominated and regular in the sense of Section 6, and all statements are local to a stratum. Singular loci are treated by stratification. Quantum, contextual, non-dominated and infinite-dimensional Gaussian experiments are treated where stated (Theorem 7.24, Proposition 9.13).

Standing scope. Four qualifications apply throughout.

1. *Canonicity is relative.* Constructions are canonical relative to the specified observation system, including its latent presentation. Two presentations with the same law image can have different liftable symmetry (Remark 5.4). The observable layer and the observable symmetry Γ_{obs} are presentation-invariant (Proposition 5.6).

2. *Descent is not selection.* Descent of a law-space structure is canonical once that structure is fixed. Its selection is a separate step (No-Go 7.3).

3. *Exact statements are conditional.* Exact-layer statements are conditional on the specified law map. Finite data enter only through confidence correspondences (Theorem 5.17), and exact law equality is not uniformly certifiable from finite samples (No-Go 8.6).

4. *Set level versus smooth level.* The kernel quotient is canonical as a set. Topological and smooth statements require H-reg.

Status of results. Each result carries one status.

- *Theorem, Proposition, Lemma*: proved here from the preceding objects.
- *Imported Theorem*: an established result, cited and not reproved.
- *Hypothesis*: a named condition H-...; a result that uses it states it.
- *No-Go*: a stronger statement is false; a counterexample is given.
- *Open Problem*: no result is asserted.

A result is called Γ-*conditional* when its hypothesis is that a stated subgroup belongs to the effective symmetry group of Definition 5.1. Such a hypothesis is tested by equality of laws.

Hypotheses used in this paper.

H-sound	declared groupoid lies in the kernel groupoid (Def. 5.8)
H-compl	declared groupoid generates every fibre (Def. 5.8)
H-coh	declared groupoid is normalized by the declared physical group (Prop. 5.11)
H-reg	regular stratum: submersive quotient, immersive descended law (Def. 6.1)
H-nuis	constant target and nuisance rank, or a declared nuisance bound (Hyp. 7.11)
H-sens	compatible sensor family (Def. 8.1)
H-top	topology and regularity for a cone boundary (Hyp. 11.2)
H-int	exact scale cocycle (Hyp. 12.1)
H-SI	SI metrological anchor, with a non-scale-covariant count-loss model (Hyp. 14.10)
H-med	propagation speed anchored or tomographically resolved (Hyp. 14.16)
H-val	held-out anchor validation (Hyp. 14.20)
H-unif	declared locally uniform convergence of a family of laws, with a modulus (Hyp. 6.7)
H-hell	locally uniform Hellinger convergence of a family (Hyp. 7.15)
H-stat	stationarity of a source law across an anchor outage (Hyp. 14.18)
H-task	declared operational task of a quantum protocol (Hyp. 7.19)
H-cov	declared latent covariance and, for dynamics, memory kernel (Hyp. 12.8)
H-resp	the system responds with the minimum-P^{-1} lift (Hyp. 12.10)
H-exch	exchangeability of a record (Hyp. 9.3)
H-mono	selection principle: metrics may not increase under Markov morphisms (Hyp. 7.4)
H-adm	declared admissible class of input declarations, with regularity (Hyp. 9.8)
H-Γ	a stated subgroup belongs to Γ (Def. 5.1)

3 Objects

Definition 3.1 (Observation system)**.** An *observation system* is a triple $\mathcal{S} = (X, \{F_i\}_{i\in I}, \Psi)$ consisting of

1. a latent space X;
2. protocols $F_i : X \to Y_i$, where Y_i is a space of observable laws (probability laws of records, or deterministic records);
3. a declared relabeling group $\Psi = \prod_i \Psi_i$, where Ψ_i acts on Y_i by pushforward along admissible bijections of outcomes.

The *joint law* is $F = (F_i)_{i \in I} : X \to Y = \prod_i Y_i$. A single experiment is the case $|I| = 1$.

Definition 3.2 (Counted channel)**.** A protocol is *counted* if its records take values in a finite chain $\{0, 1, \ldots, N\}$ or in $\mathbb{N}$, and its admissible relabelings preserve order.

Definition 3.3 (Natural module)**.** A *natural module* is a pullback $M = F^*\mu$ of a Ψ-invariant structure μ on Y: a function, a tensor field, a cone field, a family of functions with transition maps, or a transport defined from laws. A structure on X that is not of this form is *latent*.

Example 3.4. Observable invariants, clock readings, record counts and the Fisher tensor are natural. A transport generated by a latent dynamical law is latent unless it is defined from laws.

4 The exact layer

Theorem 4.1 (Kernel pair and quotient)**.** *The relation* $x \sim x' \iff F(x) = F(x')$ *is an equivalence relation. Its graph*

$$\mathcal{K} = X \times_Y X \rightrightarrows X, \qquad s(x, x') = x,\ t(x, x') = x',\ (x', x'') \circ (x, x') = (x, x''),$$

is a groupoid, the kernel groupoid*. The quotient* $\pi : X \to Q = X/\mathcal{K}$ *satisfies* $Q \cong \operatorname{im} F$ *canonically.*

Proof. Reflexivity, symmetry and transitivity are inherited from equality in Y, and these are the groupoid axioms for the graph. The map $[x] \mapsto F(x)$ is well defined and bijective onto $\operatorname{im} F$. □

Theorem 4.2 (Universal factorization)**.** *A map* $H : X \to Z$ *is constant on* $\mathcal{K}$*-orbits iff there is a unique* $\bar{H} : Q \to Z$ *with* $H = \bar{H} \circ \pi$*. In particular the observable invariants are exactly the functions on* Q*.*

Proof. Set $\bar{H}([x]) = H(x)$. It is well defined by fibre constancy and unique because π is surjective. □

Proposition 4.3 (Joint kernel)**.** $\mathcal{K} = \bigcap_i \mathcal{K}_{F_i}$*. The induced map* $Q \to \prod_i X/\mathcal{K}_{F_i}$ *is injective and in general not surjective.*

Proof. Equality in a product is equality in every factor. For non-surjectivity take $X = \{1, 2\}$, $F_1 = F_2 = \mathrm{id}$: the product of quotients has four points, Q has two. □

Proposition 4.4 (Garbling)**.** *Let* M *be a Markov kernel independent of* x*. Then* $\mathcal{K}_F \subseteq \mathcal{K}_{M \circ F}$*, with strict inclusion in general.*

Proof. $P_x = P_{x'}$ implies $MP_x = MP_{x'}$. Merging two outcomes identifies $(0.1, 0.3, 0.6)$ with $(0.3, 0.1, 0.6)$. □

No-Go 4.5 (Limits of the exact layer). The kernel groupoid and the quotient do not determine a topology, a differential structure, a metric, a statistical order, a reachability order, a cone, a scale, a connection, a clock, or a unit. Each of these is added in a later section.

5 The symmetry layer

Definition 5.1 (Stabilizer, gauge, effective symmetry)**.**

$$\mathrm{Stab} = \{(\varphi, \psi) \in \mathrm{Diff}(X) \times \Psi : \ F \circ \varphi = \psi \circ F\}, \qquad \mathrm{Gau} = \{\varphi \in \mathrm{Diff}(X) : \ F \circ \varphi = F\}.$$

Pairs whose relabelings agree on $\operatorname{im} F$ are identified. The *effective symmetry group* is $\Gamma = \mathrm{Stab}/\mathrm{Gau}$.

Theorem 5.2 (Kernel exact sequence)**.** *1. Every element of* Stab *maps* $\mathcal{K}$ *to itself:* $(\varphi \times \varphi)(\mathcal{K}) = \mathcal{K}$.

2. Gau *is a normal subgroup of* Stab.

3. Γ *acts faithfully on* Q *by* $[\varphi] \cdot [x] = [\varphi x]$*, and*

$$1 \longrightarrow \mathrm{Gau} \longrightarrow \mathrm{Stab} \longrightarrow \Gamma \longrightarrow 1$$

is exact.

Proof. If $F(x) = F(x')$ then $F(\varphi x) = \psi F(x) = \psi F(x') = F(\varphi x')$; this gives (1) and a well-defined action on Q. An element acts trivially on Q iff $F \circ \varphi = F$, i.e. iff it lies in Gau; this gives normality and faithfulness. □

No-Go 5.3 (Non-splitting). The sequence of Theorem 5.2 need not split. For $X = \mathbb{Z}_4$, $F(x) = x \bmod 2$ and the declared group $\mathbb{Z}_4$ of rotations, the gauge part is $\{0, 2\}$ and the image in $\mathrm{Aut}(Q)$ is $\mathbb{Z}_2$; the only subgroup of order two is the gauge part itself. An observable symmetry therefore need not lift to a latent symmetry of the same order.

Remark 5.4 (Liftable symmetry depends on the presentation). Γ is the effective *liftable* symmetry of the specified observation system. Equality of law images does not imply equality of Γ. Consider the Gaussian location family $F(x) = \mathcal{N}(x, 1)$ in two presentations:

- on $X_A = \mathbb{R}$, every translation lifts, so $\Gamma_A \cong (\mathbb{R}, +)$;
- on $X_B = \mathbb{R} \times \{0\} \sqcup (0, \infty) \times \{1\}$, the fibre cardinality jumps at 0, lifts must preserve it, and so $\Gamma_B = \{e\}$.

The latent presentation is an input (L0). Presentation-invariant statements require a declared equivalence of observation systems.

Definition 5.5 (Observable symmetry)**.** $\Gamma_{\mathrm{obs}} = \{\psi \in \Psi : \psi(\operatorname{im} F) = \operatorname{im} F\}$, acting on $Q \cong \operatorname{im} F$.

Proposition 5.6 (Presentation invariance of the observable layer; [67])**.** *1.* Q*, natural modules once their law-space structure is fixed, task descent, and certificates computed from data and declarations all depend on the presentation only through* $\operatorname{im} F$*. They are therefore presentation-invariant.*

2. Γ_{obs} *is presentation-invariant, and the liftable group satisfies* $\Gamma \leq \Gamma_{\mathrm{obs}}$ *(as actions on* Q*).*

3. Every statement whose proof uses the symmetry only through its action on Q *holds with* Γ_{obs} *in place of* Γ*: invariant forms and isotropy forcing, metric proportionality, and the anchor budget.*

4. Statements that use lifts (non-splitting, gauge, latent transport) remain presentation-relative.

In the Gaussian example, Γ_{obs} *contains all* 41 *tested translations in both presentations, while the liftable groups contain* 41 *and* 1.

Proof. (1) Each object is a function of law values and data. (2) The condition defining Γ_{obs} involves only $\operatorname{im} F$. An element of Stab acts on Q through ψ, which preserves the image. (3) A ψ-invariant law-space structure restricts to a Γ_{obs}-invariant structure on $\operatorname{im} F$, and the forcing arguments use only this invariance and the isotropy on T_qQ. (4) By definition. □

Lemma 5.7 (Count rigidity)**.** *The order-preserving bijections of a finite chain are the identity. Hence Ψ_i is trivial on every counted channel, and no element of* Stab *rescales a counted record.*

Proof. An order-preserving bijection of $\{0, \dots, N\}$ fixes 0 and, inductively, every successor. □

Definition 5.8 (Declared assembly)**.** A *declared groupoid* is a groupoid $A \rightrightarrows X$. It is *sound* (H-sound) if $A \subseteq \mathcal{K}$ and *complete* (H-compl) if $A \cdot x = [x]$ for all x. A *declared physical group* is a subgroup $G \leq \mathrm{Stab}$ with image $\Gamma_G \leq \Gamma$.

Theorem 5.9 (Orbit–quotient comparison)**.** *Under H-sound the map $X/A \to Q$, $[x]_A \mapsto [x]$, is a well-defined surjection. It is a bijection iff H-compl holds.*

Proof. Soundness gives well-definedness; surjectivity of π gives surjectivity; injectivity is the statement that each fibre is one A-orbit. □

No-Go 5.10 (Saturation hierarchy). The following are distinct: (a) full differential rank of a family of A-invariants; (b) global separation of A-orbits by those invariants; (c) completeness; (d) reconstruction of a latent representative. The map $(u, v) \mapsto (u^2, v)$ has full generic rank and identifies $(\pm u, v)$; a coordinate absent from F and not moved by A produces several A-orbits in one fibre; an orbit does not select a representative.

Proposition 5.11 (Coherence)**.** *If $gAg^{-1} = A$ for all $g \in G$ (H-coh), the completeness locus $\{x : A \cdot x = [x]\}$ is G-invariant. If Γ_G acts transitively on a stratum, completeness at one point of the stratum implies completeness on the stratum. Without H-coh completeness can depend on the point: on $\mathbb{Z}_6 \times \mathbb{Z}_2$ with $F(x, b) = x$, bit flips declared at even x are sound and complete exactly at even x.*

Proof. $g(A \cdot x) = (gAg^{-1})(gx) = A \cdot gx$, and $g[x] = [gx]$ by Theorem 5.2. □

5.1 The observable sector and source-side automation

For protocols $(F_i)_{i \in I}$ and a declared task T, write $F_S = (F_i)_{i \in S}$ and let $\mathcal{S}_T = \{S \subseteq I : \mathcal{K}_{F_S} \subseteq \mathcal{K}_T\}$ be the family of sufficient panels.

Theorem 5.12 (Complete minimal-panel selector; [59])**.** *Let I be a finite index set. Then $\mathcal{S}_T$ is upward closed. Its inclusion-minimal members form an antichain $\mathcal{M}_T$ with $S \in \mathcal{S}_T$ iff S contains a member of $\mathcal{M}_T$. Enumerating $\mathcal{M}_T$ is sound and complete, and it is equivariant under relabeling of states, outcomes and protocols. Ties are retained: without a declared cost, no member of $\mathcal{M}_T$ is preferred.*

Proof. Adding protocols refines the law, so $\mathcal{K}_{F_{S'}} \subseteq \mathcal{K}_{F_S}$ for $S \subseteq S'$. In a finite upward-closed family, every member contains a minimal one. Relabeling preserves equality and inclusion. □

Remark 5.13 (Infinite protocol families). Finiteness is needed. For $X = \{0, 1\}^{\mathbb{N}}$, the prefix protocols $F_n(x) = (x_1, \dots, x_n)$ and $T = \mathrm{id}$, a panel is sufficient iff it is infinite. Removing any index keeps it sufficient, so no minimal panel exists. For infinite I the upward-closed family $\mathcal{S}_T$ itself is the object.

Proposition 5.14 (No task-free isotropic selector)**.** *For $V = \mathbb{R}^m$ with isotropic covariance there is no orthogonally equivariant choice of a proper nonzero subspace. A blind spectral selector can keep a whole separated eigenspace but cannot split a degenerate one.*

Theorem 5.15 (More protocols versus more models; [59]). *Combining protocols gives* $\mathcal{K}_{(F_i)} = \bigcap_i \mathcal{K}_{F_i}$. *For an admissible model family* $\{F_\theta\}$, *the tasks observable under every model are exactly the functions constant on the equivalence closure of* $\bigcup_\theta \mathcal{K}_{F_\theta}$. *A single model-blind decoder* g *with* $T = g \circ F_\theta$ *for all* θ *exists iff* T *is constant on the equivalence closure of cross-model collisions* $\{(x, x') : F_\theta(x) = F_{\theta'}(x')\}$. *It is then unique on the union of images.*

Proposition 5.16 (Coupled law versus marginals; [60]). *For a coupled law* P^{HL} *with marginals* P^H, P^L, *we have* $\mathcal{K}_{P^{HL}} \subseteq \mathcal{K}_{P^H} \cap \mathcal{K}_{P^L}$, *with strict inclusion possible. Equality holds under a fixed coupling rule, such as declared conditional independence. Time pairing of records does not supply the coupling.*

Theorem 5.17 (Confidence-set bridge; [60]). *If* $C(y)$ *covers the true law with probability* $\geq 1-\alpha$, *then the task image* $T(\{x : P_x \in C(y)\})$ *covers* $T(x)$ *with the same probability. A point answer additionally needs a certified radius. With approximation error* η *and further gate failures of probabilities* α_g, *the error is* $\leq \tau_0 + \eta$ *with failure probability* $\leq \alpha + \sum_g \alpha_g$.

No-Go 5.18 (Tolerance is not a groupoid). The relation $d(P_x, P_{x'}) \leq \varepsilon$ is not transitive: with $\varepsilon = 1$, the points 0, 0.75 and 1.5 give two accepted adjacent pairs and a rejected endpoint pair. Its transitive closure can merge arbitrarily distant laws. Exact fibres come from exact equality or declared partitions, and approximate statements come from confidence correspondences.

6 The differential layer

Definition 6.1 (Regular stratum). A *regular stratum* is an open $\mathcal{K}$-saturated submanifold $S \subseteq X$ such that $Q_S = S/\mathcal{K}$ is a manifold, $\pi|_S$ is a submersion, and the descended law $\bar{F} : Q_S \to Y$ is an immersion into a smooth model of law space (H-reg). The *observational vertical space* is $V_x = \ker D\pi_x$.

Theorem 6.2 (Tangent sequence). *On a regular stratum*

$$0 \longrightarrow V \longrightarrow TS \xrightarrow{D\pi} \pi^* TQ_S \longrightarrow 0$$

is exact, so $T_{[x]}Q_S \cong T_x X / V_x$. *For a sound declared groupoid,* $V_x^A = T_x(A \cdot x) \subseteq V_x$, *and* V_x / V_x^A *is the infinitesimal completeness defect.*

Theorem 6.3 (Natural descent). *On a regular stratum let* $M = F^*\mu$ *be a natural module. Then*

1. $M = \pi^* \bar{M}$ *with* $\bar{M} = \bar{F}^*\mu$ *unique;* M *annihilates* V *and is independent of the representative in each fibre;*
2. $\bar{M}$ *is* Γ*-invariant;*
3. *if* M *is a symmetric tensor and the restriction of* μ *to the image tangent* $W_q = D\bar{F}_q(T_q Q)$ *is nondegenerate for every* q, *then* $\operatorname{rad} M = V$. *This holds in particular when* μ *is positive definite. In general,*
$$\operatorname{rad}(F^*\mu)_x = (D\pi_x)^{-1}\big(\operatorname{rad}(\bar{F}^*\mu)_{\pi(x)}\big).$$

Proof. (1) $F^*\mu = (\bar{F} \circ \pi)^*\mu = \pi^* \bar{F}^* \mu$. (2) For $(\varphi, \psi) \in \mathrm{Stab}$, $\varphi^* F^* \mu = (\psi \circ F)^* \mu = F^* \psi^* \mu = F^* \mu$. (3) $\bar{F}$ is immersive, so $\bar{M}_q$ is the restriction of μ to W_q. It is nondegenerate by hypothesis, and the radical of $\pi^* \bar{M}$ is $\ker D\pi$. Ambient nondegeneracy alone is not enough: for $\mu = du^2 - dv^2$ and $F(t) = (t, t)$ one has $F^*\mu = 0$ although F is an injective immersion [66]. □

Corollary 6.4 (Automatic descent). *For natural modules, basicness with respect to* $\mathcal{K}$, *representative independence, and projectability of law-defined transports and law-defined cone fields hold without further hypothesis. For latent structures they are hypotheses.*

Imported Theorem 6.5 (Singular strata; [12, 15, 53, 7]). *Orbit spaces of proper Lie groupoids carry orbit-type stratifications with slice models. At rank, isotropy or orbit-type changes the single tangent quotient is replaced by slice data and tangent cones.*

Proposition 6.6 (Tangent cone at an isotropy wall). *For $\mathbb{Z}_2$ acting by $(u, v) \mapsto (u, -v)$ and $\pi(u, v) = (u, v^2)$, $D\pi_{(0,0)}$ has rank one, while jet-rescaled secants $(\Delta u/h, \Delta r/h^2)$ fill $\mathbb{R} \times \mathbb{R}_{\geq 0}$. The first-order tangent misses a one-sided observable direction; higher jets of F are required at walls.*

6.1 Specialization across walls

A wall is reached either by a family of joint laws $F_\varepsilon \to F_0$ (a protocol wall) or along $x_n \to x$ inside one law (a latent wall). Write $\mathcal{K}_\varepsilon$, Q_ε, V_ε for the objects of F_ε.

Hypothesis 6.7 (H-unif). $F_\varepsilon \to F_0$ locally uniformly. This is all the theorems below use. For certification from data, a modulus (for instance a uniform Lipschitz bound) must additionally be declared and checked at every resolved scale, because sampling cannot see narrow features (No-Go 6.9).

Theorem 6.8 (Kernel upper semicontinuity). *Under H-unif, with F_0 continuous, $\limsup_{\varepsilon \to 0} \mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$ (Kuratowski upper limit).*

Proof. If $(x_\varepsilon, x'_\varepsilon) \in \mathcal{K}_\varepsilon$ converge to (x, x'), then $F_0(x) - F_0(x')$ is a limit of $F_\varepsilon(x_\varepsilon) - F_\varepsilon(x'_\varepsilon) = 0$, by uniform convergence and continuity of F_0. □

No-Go 6.9 (Pointwise convergence and sampling; [19]). $F_\varepsilon = \mathrm{id} + B_\varepsilon$, with a bump of height 1 and width ε^2 at $x = \varepsilon$, converges pointwise, but $(\varepsilon, 1 + \varepsilon) \in \mathcal{K}_\varepsilon$ tends to $(0, 1) \notin \mathcal{K}_0$. A sampling grid reports sup-norm 0 for this family. Locally uniform convergence is therefore declared (H-unif), not certified by sampling.

Proposition 6.10 (Specialization map).

1. *A map $\sigma : Q_\varepsilon \to Q_0$ with $\pi_0 = \sigma \circ \pi_\varepsilon$ exists iff $\mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$; it is then unique and surjective.*
2. *Every garbling wall has such a map (Proposition 4.4). Without inclusion, specialization is the correspondence $\limsup \mathcal{K}_\varepsilon$; for example, $x^2 + \varepsilon x$ identifies $(x, -x - \varepsilon)$, which x^2 separates.*
3. *Specializations compose: $\sigma_{\delta 0} \circ \sigma_{\varepsilon \delta} = \sigma_{\varepsilon 0}$.*
4. *Exactly the $\mathcal{K}_0$-basic invariants descend.*

Proof. (1) and (4) are Theorem 4.2. (3) follows from uniqueness of the factorization. □

Theorem 6.11 (Symmetry descent). *An element of $\mathrm{Stab}(F_\varepsilon)$ induces a map of Q_0 iff it maps $\mathcal{K}_0$ to itself. The descended image is a subgroup of $\mathrm{Aut}(Q_0)$ and can be a proper subgroup of Γ_0.*

Proof. $[x]_0 \mapsto [\varphi x]_0$ is well defined iff φ preserves $\mathcal{K}_0$-classes. Strictness: generic regular laws can specialize to cyclically related wall laws, so the descended image is trivial while $\Gamma_0 \supseteq \mathbb{Z}_3$ [19]. □

Theorem 6.12 (Tangent upper semicontinuity). *If $F_\varepsilon \to F_0$ in C^1 (or $x_n \to x$ within one C^1 law), every Grassmann limit point of $V_\varepsilon(x_\varepsilon)$ lies in $V_0(x)$, and $\operatorname{rank} DF_0(x) \leq \liminf \operatorname{rank} DF_\varepsilon(x_\varepsilon)$. For tame laws the rank strata are finite [37].*

Proof. For unit $v_\varepsilon \in V_\varepsilon$ with $v_\varepsilon \to v$, $DF_0 v = \lim DF_\varepsilon v_\varepsilon = 0$. Nonvanishing minors persist. □

Definition 6.13 (Stratified quotient category). The objects are the quotients of the strata. The arrows are the invertible transitions inside a stratum, together with the specializations of Proposition 6.10 (maps, or correspondences). A path is admissible if every edge is.

Theorem 6.14 (Stratified structure)**.** *Under* H-unif *the stratified quotient category is a category. A specialization is invertible iff* $\mathcal{K}_\varepsilon = \mathcal{K}_0$*. Holonomy is defined inside strata; two admissible paths into a wall are compared after specialization,* $\sigma \circ P_1$ *against* $\sigma \circ P_2$*, and differences inside the gained kernel are invisible.*

Proof. Composition of maps and of closed relations is associative. Injectivity of σ holds iff no new identifications occur. Comparison in Q_0 quotients by $\mathcal{K}_0$. □

7 The second-order layer

Definition 7.1 (Fisher module)**.** On a dominated regular experiment with score $S_u = \partial_u \log p(y \mid x)$, the Fisher tensor is $g_F(u, w) = \mathbb{E}[S_u S_w]$. It is the pullback of the Fisher–Rao metric along the law map and is a natural module.

Proposition 7.2 (Fisher descent and ignorance)**.** *On a regular stratum* $g_F = \pi^* \bar{g}_F$*,* $\bar{g}_F$ *is* Γ*-invariant, and* $\operatorname{rad} g_F = V$*. For* $Y = Jx + \varepsilon$*,* $\varepsilon \sim \mathcal{N}(0, \Sigma)$*,* $\Sigma \succ 0$*:* $g_F = J^\top \Sigma^{-1} J$ *and* $\operatorname{rad} g_F = \ker J$*.*

Proof. Theorem 6.3 with μ the Fisher–Rao metric; the Gaussian formula is standard. □

No-Go 7.3 (Naturality does not select). Descent fixes $F^*\mu$ once μ is fixed; it does not select μ. On the Bernoulli family with outcome exchange, every $a(p)\,dp^2$ with $a(p) = a(1-p) > 0$ is law-natural, relabeling-invariant and basic, and the ratios between such metrics are non-constant. Relabeling invariance therefore leaves an infinite-dimensional family [66].

Hypothesis 7.4 (H-mono)**.** *Selection principle.* A distinguishability metric may not increase under Markov morphisms (garbling): a coarser record never separates laws better. Kernel inclusion (L4) does not imply this. Merging two outcomes enlarges the kernel, yet increases Euclidean distance by up to $\sqrt{8/6}$.

Imported Theorem 7.5 (Čencov selection; [5])**.** *Under* H-mono *on finite sample spaces (congruent embeddings), the law-natural metrics are exactly* c *times Fisher,* $c > 0$*. In the test suite, Fisher is invariant to machine precision under random embeddings, while the Euclidean, quartic and* $1/p^2$ *members change by* 0.50*,* 0.53 *and* 1.00*.*

Imported Theorem 7.6 (Regular reconstruction; [49, 52])**.** *The resolved sector* $R = \operatorname{im} J$*, the null sector* $N = \ker J^*$ *and the covariance block geometry are gauge-equivariant functions of* (J, Σ, g_O)*; the Fisher tensor is the inverse Schur complement of the covariance on the resolved block.*

No-Go 7.7 (Fisher insufficiency). g_F does not determine the covariance. Covariances with equal Schur complement on the resolved block and different null blocks have the same Fisher tensor.

Proposition 7.8 (Second-order quotient; [48])**.** *Let* $O(p) \times O(q)$ *act on triples* (Σ_R, Σ_N, C)*. On the regular stratum the orbit space has dimension* $p + q + pq$*. The two spectra and the* $\min(p, q)$ *canonical correlations therefore do not coordinate it when* $pq > \min(p, q)$*; the mixed traces* $\operatorname{tr}(\Sigma_R^a C \Sigma_N^b C^\top)$ *over finite exponent ranges supply generically independent coordinates, up to a finite ambiguity.*

Proof of the dimension. Ambient dimension $\frac{p(p+1)}{2} + \frac{q(q+1)}{2} + pq$ minus group dimension $\frac{p(p-1)}{2} + \frac{q(q-1)}{2}$, with finite generic stabilizer. □

No-Go 7.9 (Shadow is not law). Distinct laws can share means and covariances (higher-moment escape), and processes with equal one-time covariance can differ in their two-time laws (memory escape). The second-order shadow is a natural module of the law, not the law.

Definition 7.10 (Efficient metric)**.** For target coordinates q and nuisance coordinates ν, $g_{\mathrm{eff}} = I_{qq} - I_{q\nu} I_{\nu\nu}^{+} I_{\nu q}$.

Hypothesis 7.11 (H-nuis)**.** Target and nuisance ranks are constant on the stratum, or a bound on the nuisance is declared and included in the efficient metric.

Proposition 7.12 (Nuisance conventions)**.** *g_{eff} is invariant under $\nu \mapsto \nu + f(q)$ and depends on the declared target chart. Without H-nuis it can be discontinuous: for $I_\varepsilon = \left(\begin{smallmatrix} a & \rho\sqrt{a\varepsilon} \\ \rho\sqrt{a\varepsilon} & \varepsilon \end{smallmatrix}\right)$, $\lim_{\varepsilon\to 0^+} g_{\mathrm{eff}} = a(1-\rho^2)$ whereas $g_{\mathrm{eff}} = a$ at $\varepsilon = 0$.*

Imported Theorem 7.13 (Data processing; [5, 1])**.** *For a parameter-independent Markov kernel, $g_{F,M\circ F} \preceq g_{F,F}$.*

No-Go 7.14 (Kernel order is not Blackwell order; [3, 11]). Two injective experiments have the same kernel (the diagonal) but need not be Blackwell equivalent: $\mathcal{N}(x, 1)$ and $\mathcal{N}(x, 4)$ are both injective and the second is a strict garbling of the first.

Hypothesis 7.15 (H-hell)**.** A family $F_\varepsilon \to F_0$ is locally uniformly Hellinger-continuous; for dominated regular families with uniformly differentiable square-root densities this is Le Cam convergence of the local experiments [38].

Theorem 7.16 (Specialization of distinguishability)**.** *Under H-unif and H-hell, Hellinger distances, natural Fisher modules and Bayes risks converge to their wall values,* $\operatorname{rad} g_0 \supseteq \limsup V_\varepsilon$*, and the Cramér–Rao bound diverges on the gained kernel.*

Proof. The Hellinger triangle inequality gives distances; the second-order Hellinger expansion gives Fisher; total variation is bounded by Hellinger distance, which gives risks. □

No-Go 7.17 (Weak convergence is not enough). For y observed as $\mathcal{N}(\varepsilon y, \varepsilon^2)$, the laws converge weakly to a wall law identifying y. Yet the Fisher information in y stays 1, the Hellinger distance between $y = 0.3$ and $y' = 1.1$ stays 0.277, and the minimal error of testing $y = 0$ against $y = 1$ stays 0.309 instead of $1/2$.

7.1 The quantum metric module

For finite-dimensional quantum protocols, $F(\theta)$ is the joint outcome law of a declared measurement family on ρ_θ. When the family is tomographically complete, the kernel is equality of states. Natural metric modules must be monotone under CPTP maps.

Imported Theorem 7.18 (Petz classification; [26])**.** *The CPTP-monotone Riemannian metrics on full-rank states are*

$$c\, g_f(\rho; A, A) = c \sum_{i,j} \frac{|A_{ij}|^2}{\lambda_j f(\lambda_i/\lambda_j)},$$

with f operator monotone, $f(t) = tf(1/t)$, $f(1) = 1$ and $c > 0$. On commuting tasks all g_f equal the classical Fisher metric. Monotonicity is therefore a family gate, not a selector.

Hypothesis 7.19 (H-task)**.** The quantum protocol declares its operational task (estimation, asymmetric testing, symmetric testing, or another declared local task).

Theorem 7.20 (Task lock; [27, 28, 29, 30, 31, 20])**.** *Under H-task, the task fixes the Petz member up to scale:*

- *optimal-measurement estimation gives SLD, since* $\max_{\mathrm{POVM}} F_{\mathrm{cl}} = g_{\mathrm{SLD}}$*;*
- *Stein testing gives BKM, the Hessian of the Umegaki relative entropy;*
- *Chernoff testing gives Wigner–Yanase, since $\xi = h^2 g_{\mathrm{WY}}/8$ locally.*

Theorem 7.21 (Nonparametric identification; [20]). *Commuting tasks fix the scale. For* $\rho = \mathrm{diag}(\lambda_1, \lambda_2)$ *and* $A = \sigma_x/\sqrt{2}$*, a response* y *determines* $f(\lambda_1/\lambda_2) = 2\kappa|A_{12}|^2/(\lambda_2 y)$*. Dense ratios determine* f*. Finite-panel fits are not locks; a metric certificate is a band of* f *of width set by noise divided by the response gap.*

Theorem 7.22 (Quantum wall specialization; [20]). *On tangents coupling support and kernel at a support wall,* g_f *specializes iff* $f(0) > 0$*. SLD tends to* $4\times$ *Fubini–Study. BKM, Heinz*$(1/4)$ *and RLD diverge like* $\log(1/\varepsilon)$*,* $\varepsilon^{-1/4}$ *and* ε^{-1}*.*

Wall geometry is obtained by re-discrimination of the specialized task, with outcomes LOCKED, DIVERGENT, VACUOUS, UNDERDETERMINED and INCONSISTENT. Metric selection selects no transport.

Imported Theorem 7.23 (Trace rule; [70, 71]). *Noncontextual probabilities on the projections of a Hilbert space of dimension* ≥ 3*, and on effects in every dimension, are* $\mathrm{Tr}(\rho E)$*. A squared rule is not normalized across bases (basis dependence* 0.20 *in the test suite).*

Theorem 7.24 (Contextual protocols; [72, 73, 62]).

- *No assignment of noncontextual values reproduces quantum observables:* 0 *of the* 512 *assignments of* ± 1 *satisfy the Peres–Mermin square.*
- *A coupled law over the contexts of a CHSH scenario exists iff all CHSH inequalities hold (Fine). The quantum box with value* $2\sqrt{2}$ *admits none; linear-programming feasibility agrees with Fine's criterion on* $60/60$ *random boxes.*

Hence, for incompatible measurement families, the product-valued law map over contexts is the correct observation law. Its kernel, quotient and task-selected Petz layer apply unchanged, and no coupled-law rung is required. Quantum uncertainty is not ignorance over a latent kernel of hidden values.

8 Ignorance and the observation limit

Definition 8.1 (Persistent null). For a family $\{F_\rho\}$ of protocols on a common latent space (H-sens), the persistent null at x is $\mathcal{I}_x = \bigcap_\rho \ker DF_\rho|_x$.

Proposition 8.2 (Finite attainment). *In finite dimension* $\mathcal{I}_x$ *equals the intersection over finitely many members of the family, and it equals* V_x *of the joint law of that finite subfamily on its regular stratum.*

Proof. A descending chain of subspaces of a finite-dimensional space stabilizes. The joint statement is Proposition 4.3 at the tangent level. □

Proposition 8.3 (Three layers). *For smooth maps* E *(evaluation at finite precision) and* R *(downstream representation), the exact null of the sensor is* $\ker DF$*. The precision loss of* E *and the representation loss of* R *are nulls of the composites* $E \circ F$ *and* $R \circ E \circ F$ *and satisfy* $\ker DF \subseteq \ker D(E \circ F) \subseteq \ker D(R \circ E \circ F)$*. Only* $\ker DF$ *is a property of the sensor.*

Imported Theorem 8.4 (Finite-sample boundary reconstruction; [47]). *For a correctly specified regular protocol with a design that places positive mass on both sides of an operational boundary and a nonsingular Fisher matrix, the maximum-likelihood boundary estimator is consistent and asymptotically normal with covariance given by the inverse Fisher matrix. The quotient alone determines no numerical boundary: affine recalibration of the latent coordinate moves it.*

Proposition 8.5 (Rank walls are not certified from finite data). *A rank certificate is FULL when a lower confidence bound of* $\sigma_{\min}$ *exceeds a declared* τ *at sample sizes* n *and* $4n$*; otherwise only an upper bound on* $\sigma_{\min}$ *is reported. A true wall and a nearby full-rank law with* $\sigma_{\min} < \tau$ *give the same certificate [19].*

No-Go 8.6 (Exact law equality is not uniformly certifiable). Take $P_a = \text{Bernoulli}(\frac{1}{2})$, with $P_b = P_a$ under one model and $P_b = \text{Bernoulli}(\frac{1}{2} + \delta)$ under the other. The two exact kernels differ for every $\delta \neq 0$, yet the n-sample laws satisfy $\text{TV} \leq n|\delta|$. With $\delta = 10^{-6}$ and $n = 10^4$, a level-0.05 test rejects at rate 0.06. Kernel statements about real data are therefore set-valued, through confidence correspondences, unless a separation margin is declared [66].

9 Uncertainty, probability and forgetting

Definition 9.1 (Primitive of uncertainty)**.** The uncertainty of a declared experiment is its indistinguishability:

- the kernel $\mathcal{K}$ at the exact layer;
- the non-transitive tolerance $H(F(x), F(x')) \leq \varepsilon$ at finite resolution;
- the Fisher metric infinitesimally, selected under H-mono (Imported Theorem 7.5).

Theorem 9.2 (Uncertainty splits along the quotient; [57])**.** *The component of a requested quantity that factors through Q is constrained by data. Its component along the fibres of $\mathcal{K}$ never is. In a regular model, posteriors under two priors merge on the quotient (Bernstein–von Mises [74]). Along the kernel, the posterior equals the conditional prior on the fibre for every sample size.*

Proof. The likelihood is constant on fibres (Theorem 4.2), so Bayes' rule multiplies the prior by a function of $\pi(x)$. Bernstein–von Mises applies on the quotient. □

Hypothesis 9.3 (H-exch)**.** The record is exchangeable.

Imported Theorem 9.4 (de Finetti; [69])**.** *Under H-exch a record is a mixture of i.i.d. laws, and its frequencies converge. Without it, no probability is forced: a deterministic record of doubling blocks oscillates forever.*

Theorem 9.5 (Gaussian propagation is the LAN limit; [38, 57])**.** *On a regular stratum, linearized error propagation is the locally asymptotically normal limit (coverage 0.947 at nominal 0.95). Across a fold, where two branches give the same law, it selects one branch (coverage 0 on the other). The preimage correspondence keeps nominal coverage.*

Forgetting of declarations. Write $D_t(\mathcal{A}) = \sup_{\mu,\nu \in \mathcal{A}} d(\Phi_{t\#}\mu, \Phi_{t\#}\nu)$ for the declaration diameter of an admissible class $\mathcal{A}$ [58].

No-Go 9.6 (TV rigidity under bijections). For a bimeasurable bijection Φ, $\text{TV}(\Phi_\#\mu, \Phi_\#\nu) = \text{TV}(\mu, \nu)$. Invertible dynamics (ODE flows, Galerkin Navier–Stokes, shell models) never forget in total variation on the full state. Forgetting there happens only on observation fibres (fibre migration, which can be non-monotone), in a weak metric, or with noise.

No-Go 9.7 (No uniform forgetting without a declared class). If $\mathcal{A}$ contains uniform laws on all small balls and the observed flow is continuous and non-constant, then $D_t^{\text{TV}}(\mathcal{A}) = 1$ for every t, even for exact maps. Finite lists of declarations cannot stand in for a class. In the test suite, concentrating laws keep $\text{TV} \geq 0.75$ at every fixed n.

Hypothesis 9.8 (H-adm)**.** The admissible class of input declarations is declared before the output is seen, with a regularity that makes a supremum finite. Examples are a ball in bounded variation, an L^1-compact set, an energy ball, or a band limit.

Theorem 9.9 (Mixing locks; [75, 76, 58])**.** *1. If the transfer operator is exact and $\mathcal{A}$ is L^1-compact, then $D_n^{\text{TV}}(\mathcal{A}) \to 0$.*

2. *For the tent map and densities of bounded variation,*

$$\mathrm{TV}(T^n_{\#}\mu_f, T^n_{\#}\mu_g) \le 2^{-(n+2)}\mathrm{Var}(f-g),$$

with equality for every n at $f = 2^{n+1}\mathbf{1}_{[0,2^{-n-1}]}$ and $g = 1$. By the conjugacy $h(x) = \sin^2(\pi x/2)$ the same bound holds for the logistic map at $r = 4$. In exact rationals the largest random ratio is 0.0099 *and the witness attains* 1.

Theorem 9.10 (Contraction, noise and functional type; [58, 77, 78])**.**

- *For exact-dealiased 2D Galerkin Navier–Stokes on an energy ball, $W_1(\Phi_{t\#}\mu, \Phi_{t\#}\nu) \le \Lambda(t)W_1(\mu,\nu)$, with forgetting certified eventually iff $G_N = S\|f\|/\nu^2 < 1$.*
- *For Markov kernels, $\mathrm{TV}(\mu K^n, \nu K^n) \le \delta(K^m)^{\lfloor n/m \rfloor}$ (Dobrushin). Deterministic kernels have $\delta = 1$.*
- *Time averages of uniquely ergodic systems forget every declaration, including Dirac masses, at explicit rates. Fixed-time laws need mixing on a class. The functional type is part of the declaration.*

No-Go 9.11 (Samples refute, operators lock; [11, 79])*.* Without a regularity class, no finite-sample procedure certifies TV $< \frac{1}{2}$: for laws uniform on M cells against uniform on a random half, any test from m draws has power $\le \alpha + 1.5m^2/M$. A lock is an inequality on the transfer operator, flow or kernel; samples serve to refute it. Agreement of two ensembles does not certify accuracy: floating-point orbits are atomic and can be wrong together.

Theorem 9.12 (Non-circularity certificate; [57, 58])**.** *Under* H-adm*, an uncertainty statement is FORCED at scale τ from time t^* iff $D_t(\mathcal{A}) \le \tau$ for all $t \ge t^*$, certified by an operator inequality; otherwise it is CONDITIONAL and reported with its declaration. A FORCED statement contains no input dependence beyond τ, so nothing declared is smuggled. Residual dependence is supported on the observational kernel (Theorem 9.2), before mixing, or on the invariant weights of non-mixing dynamics. Monte Carlo is the integration engine for FORCED targets. It carries its own effective-sample-size certificate, and it is not the lock.*

Proposition 9.13 (Non-dominated and infinite-dimensional strata)**.** *(See [80, 81].)*

- *The exact, total-variation and Hellinger layers need no global dominating measure, since every pair is dominated by its sum. A Dirac location family is perfectly distinguishable at every shift ($TV = 1$), and only its Fisher layer is absent.*
- *Gaussian measures on infinite-dimensional spaces are either equivalent or mutually singular (Feldman–Hájek). Shifts in the Cameron–Martin space keep $TV < 1$ (it converges to* 0.25 *in the test); others become perfectly distinguishable ($TV \to 1$).*

The Fisher layer of a Gaussian field or path-space experiment is the Cameron–Martin geometry. Every other direction is resolved perfectly, as a singular stratum.

10 Statistical distance and angle

Definition 10.1 (RG length, distance, angle)**.** On a regular stratum on which $\bar{g}_{\mathrm{eff}}$ is positive definite,

$$L[\gamma] = \int \sqrt{\bar{g}_{\mathrm{eff}}(\dot\gamma,\dot\gamma)}\,dt, \quad d_{RG}(q_0,q_1) = \inf_\gamma L[\gamma], \quad \cos\angle(u,w) = \frac{\bar{g}_{\mathrm{eff}}(u,w)}{\|u\|\,\|w\|}.$$

Theorem 10.2 (Distance law)**.** *Under* H-reg *and* H-nuis*, d_{RG} is the Riemannian distance of $\bar{g}_{\mathrm{eff}}$ on Q_S. It is independent of the latent representative and of the latent chart, Γ-invariant, and dimensionless.*

Proof. Theorem 6.3 and Proposition 7.2; the Fisher tensor is invariant under reparametrization and the score is dimensionless. □

Proposition 10.3 (Monotonicity)**.** *Under a parameter-independent garbling, path lengths and d_{RG} do not increase.*

No-Go 10.4 (Rank one). On a rank-one stratum, length and order along the resolved direction are defined, and no angle between independent directions is.

No-Go 10.5 (Finite divergences). Only the intrinsic length is additive under concatenation. For the Gaussian location family the Bhattacharyya angle between p and $p(\cdot - a)$ is $2 \arccos e^{-a^2/8\sigma^2}$, which is not linear in a and is bounded by π.

No-Go 10.6 (Experiment relativity). For N samples of $\mathcal{N}(x, \sigma^2)$, $d_{RG} = \sqrt{N}\,|\Delta x|/\sigma$. On a class of experiments closed under additive-noise garbling there is no function f with $f(d_{RG}) = |\Delta x|$: $d_{RG}(\Delta x = 1, \tau = 0) = d_{RG}(\Delta x = 2, \tau = \sqrt{3}\sigma)$ for added noise of variance τ^2.

11 Order, cone and conformal class

Definition 11.1 (Reachability)**.** A declared preorder $\preceq$ on X is *fibre-compatible* if it is constant on $\mathcal{K}$-orbits. It descends to Q; its antisymmetrization is the causal order.

Hypothesis 11.2 (H-top)**.** Q_S carries a topology compatible with $\preceq$, and the future sets have a regular boundary whose tangent directions form a cone field $C_q \subset T_q Q_S$.

No-Go 11.3 (Projective typing). The realization type of C_q is determined by its projective boundary, not by a defining function: $\{Q = 0\}$ and $\{Q^3 = 0\}$ define the same cone.

Theorem 11.4 (Realization branches)**.** *Under H-top the first applicable case describes C_q on a stratum:*

1. *the null cone of a nondegenerate Lorentzian quadratic form: then C determines a positive conformal ray $[g]$ (Imported Theorem 11.5);*
2. *a single smooth, strongly convex, homogeneous cone with Lorentzian fibre Hessian: a Lorentz–Finsler structure;*
3. *several persistent characteristic sheets: a multicone or multi-Finsler structure;*
4. *a cone whose rank, signature, topology, isotropy or sheet count changes: stratified realization by slices and tangent cones (Imported Theorem 6.5);*
5. *otherwise: general characteristic or admissibility geometry.*

Imported Theorem 11.5 (Conformal reconstruction; [8, 13])**.** *Two nondegenerate Lorentzian quadratic forms on a real vector space with the same null cone are proportional; a time orientation fixes the sign. Globally, under distinguishing causality, the causal structure of a Lorentzian manifold determines its conformal class and not its conformal factor.*

Theorem 11.6 (Lorentz isotropy forces the quadratic branch)**.** *Suppose the isotropy Γ_q acts on $T_q Q \cong \mathbb{R}^{1+n}$ and contains $SO^+(1, n)$ in its standard representation, $n \geq 2$ (H-Γ). Then every Γ_q-invariant closed convex pointed cone with nonempty interior is the future or the past light cone, and the Γ_q-invariant symmetric bilinear forms are $\{\lambda\eta : \lambda \in \mathbb{R}\}$. For $n = 1$ the conclusion fails: the wedge $\{x \geq |t|\}$ is invariant.*

Proof. An invariant cone is a union of orbits: sheets of two-sheeted hyperboloids, null half-cones, one-sheeted hyperboloids and the origin. For $n \geq 2$ a one-sheeted hyperboloid contains v and $-v$ and is excluded by pointedness. Invariance under positive scaling, convexity and closedness leave one closed light cone. The standard representation is absolutely irreducible, so invariant symmetric forms form a line. □

11.1 The residual characteristic branch

Theorem 11.7 (Tame reduction; [44, 37, 18])**.** *If $\mathbb{P}A$ is definable in an o-minimal structure, then:*

- *it has a finite canonical Whitney stratification with constant signature of the second fundamental form on each stratum;*
- *it has finitely many sheets ($\leq (d-1)(d-2)/2+1$ for a degree-d plane curve, Harnack) and integer dimension;*
- *its finite-resolution component counts stabilize;*
- *its type, consisting of the strata, their incidences and signatures, and the sheet monodromy, is finite combinatorial data. Bounded-complexity families have finitely many types up to stratified homeomorphism.*

No-Go 11.8 (Linear moduli). Polyhedral cones with five extreme rays carry continuous projective invariants. Tame types are classified up to stratified homeomorphism, not up to $GL(V)$.

Imported Theorem 11.9 (Duality; [42])**.** *A closed convex pointed cone is smooth iff its dual is strictly convex. A Lorentz–Finsler realization requires a primal and a dual that are both smooth.*

Theorem 11.10 (Hyperbolic lock; [35, 36, 41, 18])**.** *If the characteristic set is the zero set of a principal symbol hyperbolic in a time direction (for constant coefficients, a well-posed Cauchy problem), then:*

- *the forward cone is convex;*
- *the sheets are real, ordered and nested, with trivial label monodromy;*
- *the only residual structure is contact between sheets, with $\mathbb{Z}_2$ polarization holonomy at conical points.*

Conversely, a nonconvex or label-exchanging characteristic set of a principal symbol forces complex roots, with growth rate proportional to frequency (Hadamard ill-posedness).

No-Go 11.11 (Wild class). Non-definable direction sets (fractal, infinitely accumulating) are characterized only by resolution invariants: count exponent, box dimension, Cantor–Bendixson rank. They admit no finite classification, and finite data certify only window statements: no finite window certifies tameness or wildness [18].

12 Scale and transport

Hypothesis 12.1 (H-int)**.** Local scale representatives $g_i = \Omega_i^2 g$ on a cover differ on overlaps by positive functions $\rho_{ij} = \Omega_i/\Omega_j$ forming an exact cocycle, $\rho_{ij} = \sigma_i/\sigma_j$.

Proposition 12.2 (Scale representative)**.** *A global metric representative of $[g]$ exists on a connected stratum iff H-int holds. A nonzero loop product of ρ is scale holonomy: a unit transported around the loop returns multiplied by it. Data along one worldline fix at most one normalization along that worldline.*

Proposition 12.3 (Strict transport)**.** *If two representatives g and $\Omega^2 g$ are both preserved by a strict transport on a transport-connected stratum, Ω is constant there.*

Proof. $P_{qr}^*(\Omega^2 g)_r = (\Omega^2 g)_q$ and $P_{qr}^* g_r = g_q$ give $\Omega(r) = \Omega(q)$ by nondegeneracy. □

Imported Theorem 12.4 (Induced transport)**.** *On a regular embedded metric stratum with orthogonal projector P, the induced connection is $\nabla^W = P\nabla^{LC}$ and the second fundamental form is $(I-P)\nabla^{LC}$.*

No-Go 12.5 (Entropy does not select a connection). $H(X)$, $H(Q)$ and $H(X \mid Q)$ are invariant under fibre-preserving automorphisms, and horizontal lifts are not; no function of these entropies selects a lift.

Proposition 12.6 (Law-defined transport)**.** *A transport defined from laws descends uniquely to Q_S (Corollary 6.4). A latent transport W descends iff $D\pi(W_x)$ is constant on fibres.*

12.1 Canonical transport

Theorem 12.7 (No law-natural connection; [21])**.** *For lifts B_1, B_2 with $DF\,B_i = I$, the vertical shear $\varphi = \mathrm{id} + (B_2 - B_1)DF$ preserves the law and maps B_1 to B_2. Hence* Gau *acts transitively on horizontal complements, and no* Gau*-invariant connection exists when $V \neq 0$. Laws, natural modules, entropy, wall specialization and quantum metric selection therefore select no connection.*

Proof. $DF(B_2 - B_1) = 0$, and $\varphi B_1 = B_1 + (B_2 - B_1)DF\,B_1 = B_2$. Natural modules are Gau-invariant (Theorem 6.3). □

Hypothesis 12.8 (H-cov)**.** A latent covariance $P \succ 0$ is declared, and for dynamics also a memory kernel (a latent process law).

Theorem 12.9 (Canonical connection; [21, 32, 33, 34])**.** *Under H-cov, $H_P = \mathrm{im}(P\,DF^\top) = V^{\perp_{P^{-1}}}$ is a complement to V, and:*

1. *it is the unique minimal-P^{-1}-norm lift and the noiseless limit of the Kalman update;*
2. *it is equivariant, and different declared P give different transports on identical laws;*
3. *its curvature (the O'Neill tensor) is the latent holonomy of closed observed loops, of order their area, and zero for linear laws;*
4. *with a memory kernel, the canonical lift is the path-space conditional mean, which equals the RTS smoother;*
5. *across a rank wall with the same P, $H_{P,0} \subseteq \lim H_{P,\varepsilon}$, and wall transport is the P^{-1}-orthogonal projection onto $H_{P,0}$.*

Proof. (1) Minimal-norm solution of $DF\,x = \delta y$; let $R \to 0$ in the Kalman gain. (2) Substitution. (3) O'Neill's integrability tensor. (4) Gaussian conditioning on path space. (5) Tangent upper semicontinuity (Theorem 6.12) and reversal of inclusions under complements. □

Hypothesis 12.10 (H-resp)**.** The physical system responds to a requested observed velocity with the minimum-P^{-1} lift.

Proposition 12.11 (The canonical connection as a law of motion; [63, 64, 65])**.** *H-resp holds exactly in two classes of systems:*

- *overdamped systems with no external force on the unobserved directions, where P is the mobility;*
- *free bodies whose momentum conjugate to the unobserved (symmetry) directions is conserved and zero, where P is the inverse inertia.*

There the curvature of Theorem 12.9 is physical, and a closed observed loop moves the unobserved configuration by oriented area times curvature. Examples are the three-sphere Stokes swimmer, with curvature $(a/3)[L_1^{-2} + L_2^{-2} - (L_1 + L_2)^{-2}]$, and the falling-cat reorientation. Reciprocal strokes give zero (Purcell's scallop theorem, as flatness). By the Einstein relation, and because B is invariant under $P \to cP$, the estimation connection fitted to passive fluctuations equals the physical one, and it predicts active locomotion (to within 1–1.5% in simulation).

13 Duration

Let $T_i : H \to \mathbb{R}$ be the readouts of clock protocols on a history space H, i.e. natural modules of the clock channels.

Theorem 13.1 (Clock transition groupoid)**.** *If $\mathcal{K}_{T_i} = \mathcal{K}_{T_j}$, there is a unique bijection $f_{ji} : T_i(H) \to T_j(H)$ with $T_j = f_{ji} \circ T_i$, and*

$$f_{ii} = \mathrm{id}, \qquad f_{ij} = f_{ji}^{-1}, \qquad f_{ki} = f_{kj} \circ f_{ji}.$$

If $\mathcal{K}_{T_i} \neq \mathcal{K}_{T_j}$ no such map exists.

Proof. Theorem 4.2 applied to the common quotient; uniqueness closes composition. If the kernels differ, some pair is identified by one readout and separated by the other. □

Theorem 13.2 (Affine clocks)**.** *Suppose a flow Φ_s belongs to* Stab*, $T_i \circ \Phi_s = T_i + a_i s$ with $a_i > 0$, and each $T_i(H)$ is an interval (H-Γ). Then every f_{ji} is affine with slope a_j/a_i. The clocks are charts of one duration object, unique up to a positive unit and an origin on each connected component.*

Proof. $f_{ji}(T_i \circ \Phi_s) = T_j \circ \Phi_s$ gives $f_{ji}(t + a_i s) = f_{ji}(t) + a_j s$ for all admissible t, s. □

Proposition 13.3 (Path independence)**.** *An accumulated duration T is a function of a declared increment Δt iff $\mathcal{K}_{\Delta t} \subseteq \mathcal{K}_T$ on H. Unsigned elapsed duration is invariant under path reversal and fixes no time orientation.*

Imported Theorem 13.4 (Coherence duration; [16])**.** *For multiplicative coherence operators C, $\mathcal{T} = -\log C$ is additive on commuting families, and the defect of additivity is the Baker–Campbell–Hausdorff series $\frac{1}{2}[\mathcal{T}_1, \mathcal{T}_2] + \cdots$.*

13.1 Dynamics

For a map f with a finite observation s, the history protocols are $F_n = (s, s \circ f, \ldots, s \circ f^{n-1})$. Time translation acts on records by the shift.

Proposition 13.5 (Finite-history regularity; [17])**.** *For the logistic family with $x_0 \sim U(0,1)$, the n-word law is continuous in r and smooth off finitely many kneading walls, with finite Fisher–Rao length across them. The noiseless wall (non-domination, discontinuous law map) occurs only as $n \to \infty$.*

Imported Theorem 13.6 (Kernel-count entropy; [40, 17])**.** *For a unimodal f and its critical partition, $|X/\mathcal{K}_{F_n}|$ is the lap number of f^n, and $h_{\mathrm{top}} = \lim \frac{1}{n} \log |X/\mathcal{K}_{F_n}| = \inf_n \frac{1}{n} \log |X/\mathcal{K}_{F_n}|$.*

Imported Theorem 13.7 (Entropy locks; [43, 39])**.** *For ergodic μ of a C^2 interval map, $h_\mu \leq \max(\lambda, 0) \leq h_{\mathrm{top}}$, with $h_\mu = \lambda$ iff μ is absolutely continuous.*

Definition 13.8 (Typed certificate)**.** A dynamical certificate names its module (kernel count, history law, derivative, exact return, resolution) and its escalation window. It is issued only on an escalation plateau and only if it is consistent with Imported Theorem 13.7 and with exact return for periodicity. Zero entropy and a rank wall are never certified from finite data.

Imported Theorem 13.9 (Cone-field criterion; [46])**.** *A continuous strictly invariant cone field with uniform expansion exists iff the dynamics are uniformly hyperbolic. Nonuniformly hyperbolic characteristic processes are typed per window.*

13.2 Chaos is not hidden: the frontier law

Theorem 13.10 (Frontier law; [83, 56])**.** *For an ergodic system with top exponent $\lambda > 0$, let a forecast of accuracy ε and coverage q be declared, from an initial measurement of precision σ_0. The largest certified (LOCKED) horizon is*

$$n_q^*(\sigma_0) = \frac{\log(\varepsilon/\sigma_0)}{\lambda_q(n^*)} + c_P + o(1),$$

where λ_q is the q-quantile of the finite-time exponent and c_P depends only on the protocol. The slope in $\log(1/\sigma_0)$ tends to $1/\lambda$, and protocols move only the intercept. If $\lambda < 0$ the frontier is infinite below the transient scale.

Proof. Linearize, $\log|\delta_n| = \log\sigma_0 + \log|Z_P| + n\lambda_n$. The protocol changes only the law of Z_P. Finite-time exponents converge to λ (Oseledets). □

Recovered exponents across five seeds:

- logistic $r = 4$: 0.674–0.696 (true 0.693);
- logistic $r = 3.9$, median frontier: 0.496–0.509 (true 0.496);
- Hénon: 0.414–0.421 (true 0.420);
- Lorenz: 0.87–0.93 (true 0.906).

Slopes across three sensor protocols agree within 1–7%. The frontier disappears only if the task is changed, for example by re-observing the state (filtering), which is a declaration.

Corollary 13.11 (Duality and quotient growth)**.** *Fisher information about the initial state grows like $e^{2\lambda n}$, so the past becomes resolvable at twice the rate at which the future is lost (measured 1.015 against $2\lambda = 0.993$). For a generating partition, the observable quotient of n-step histories has $\log|Q_n| = hn + c$, where c depends on the resolution. At 2, 4 and 8 bins the measured slopes are 0.6931, 0.6931 and 0.6950, against $\ln 2$.*

Theorem 13.12 (Information cost of chaos; [84, 85])**.** *For almost every orbit, the algorithmic complexity of its symbolic itinerary per step equals the Kolmogorov–Sinai entropy (Brudno). For SRB measures this entropy is the sum of positive exponents (Pesin). Each certified forecast step therefore costs $\lambda/\ln 2$ bits of initial precision. Compression upper bounds measured in the test suite: 1.00 bit per step at $r = 4$, 0.82 at $r = 3.9$ (entropy 0.72), and 0.0014 for a periodic orbit.*

Proposition 13.13 (Clock distortion classes; [61])**.** *Let perceived time be $\tau = w(t)$. Then:*

- *an offset $w = t + c$ preserves all differences;*
- *a rate $w = kt + c$ preserves all ratios and angles but no distance;*
- *a monotone warp preserves only order and counts.*

Under a log-rate random walk, a distance calibrated at time 0 has log-error growing like $\sqrt{T}$ (measured exponents 0.494–0.510), while angles from delays measured together stay exact. A common rate is a dilation gauge, broken only by a second independent clock.

14 Physical distance

14.1 No internal unit

No-Go 14.1 (Unit).

1. If a readout scale s is not anchored, the law depends on $(x/s, \sigma/s)$, and the dilation $(x, \sigma, s) \mapsto (kx, k\sigma, ks)$ lies in Gau.
2. Markov-invariant metrics on finite simplices are c times the Fisher metric with $c > 0$ free [5].
3. The set $\{k_B, T, m\}$ contains no quantity of dimension length; a length requires an action or a pair of clock frequency and speed [4]. Thermodynamic length is invariant under a change of energy unit [6].
4. A flat metric is Euclidean in some chart; the chart need not be the physical one.

Proposition 14.2 (Counted channel removes dilation)**.** *If F contains a counted channel whose law changes under a physical dilation, that dilation is not in* Stab*. If F contains only continuous channels whose relabeling groups include rescaling, dilations may lie in* Stab*.*

Proof. Lemma 5.7: no relabeling compensates a changed count law. □

No-Go 14.3 (Scale-covariant count loss; [17])*.* If the anchor channel loses counts multiplicatively, $n' = (1-\epsilon)n$ with ϵ unknown, the dilation fibre returns: Fisher rank 1 in (scale, ϵ). Additive slips and rate-dependent dead time do not reopen it. H-SI therefore includes a non-scale-covariant loss model.

14.2 Displacement and invariant metric

Definition 14.4 (Displacement)**.** A displacement is an element of Γ acting on a target chart; concatenation is composition.

Imported Theorem 14.5 (Ratio scale; [9, 10])**.** *A monotone additive representation of concatenation is linear and unique up to one positive constant.*

Theorem 14.6 (Two-point invariant)**.** *Suppose the latent space contains station and target positions in $\mathbb{R}^n$ and* Gau *contains the diagonal action of $SE(n)$, $n \geq 2$, or of $E(n)$, $n \geq 1$ (H-Γ). Then the orbit space of station–target pairs is $[0, \infty)$, parametrized by Euclidean separation ρ, and every natural two-point scalar is a function of ρ.*

Proof. The group is transitive on pairs at fixed separation: rotate (or, for $n = 1$, reflect) the difference vector, then translate. An invariant function is constant on orbits. □

Theorem 14.7 (Metric proportionality)**.** *Suppose relative Euclidean motions of the target lie in Γ, act transitively on the target chart, and have isotropy acting irreducibly on the tangent space (H-Γ). Then every natural metric module on the chart satisfies $\bar{g} = \lambda^2 h$, where h is the Euclidean metric of ρ and $\lambda > 0$ is constant.*

Proof. $\bar{g}$ and h are Γ-invariant (Theorem 6.3). The invariant symmetric forms of an irreducible compact isotropy form a line, since the commutant is $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$ and its symmetric part is $\mathbb{R}$; so $\bar{g} = \lambda(q)^2 h$. The ratio λ is a Γ-invariant function and Γ is transitive. □

Theorem 14.8 (Anchor budget; [17])**.** *If the empirical isotropy is K, a constant metric on the target chart is fixed by exactly* $\dim(\mathrm{Sym}^2 V^*)^K$ *radar anchors in general position:* 1 *under irreducible isotropy,* $n(n+1)/2$ *without symmetry.*

Proof. The admissible metrics form the linear space $(\mathrm{Sym}^2 V^*)^K$, and each anchor is one linear equation. □

Imported Theorem 14.9 (Unit-distance rigidity; [2]). *For $n \geq 2$, a map $\mathbb{R}^n \to \mathbb{R}^n$ preserving distance one is an isometry. For $n = 1$ this fails: $x \mapsto x + 0.1 \sin 2\pi x$ preserves distance one and not distance one half.*

14.3 Anchor and identification

Hypothesis 14.10 (H-SI). The slice contains a two-way timing protocol that counts caesium cycles; c and $\Delta\nu_{\mathrm{Cs}}$ take their exact SI values [51]; and round-trip counts add along concatenations for which ρ attains equality in the triangle inequality.

Proposition 14.11 (Radar distance). *Under H-SI the round-trip count n_{Cs} determines*

$$d_{\mathrm{rad}} = \frac{c}{2\Delta\nu_{\mathrm{Cs}}}\, n_{\mathrm{Cs}},$$

and the factor $\frac{1}{2}$ is independent of the one-way synchronization convention [50]: with one-way speeds $c/(2\varepsilon)$ and $c/(2(1-\varepsilon))$ the round-trip time does not depend on ε.

Proposition 14.12 (Radar and rod distance). *On a static slice with metric representative $W^2(-dt^2 + dx^2)$ and a clock with $W = 1$ at the station, radar distance is independent of W away from the station, while rod distance is $\int W\, dx$. They agree on a path iff the exchanged-frequency ratio $W(x)/W(0)$ equals one along it. If translations along the path act in Γ on the clock channel, this ratio equals one.*

Proof. Null curves are conformally invariant, so the round-trip proper time at the station does not depend on W elsewhere. If a translation is in Stab, it maps the clock law at x to the clock law at $x + a$ with trivial relabeling on counts (Lemma 5.7), so the counted rates agree. □

Imported Theorem 14.13 (Stationary slices; [23, 24]). *On a stationary slice with metric $g_{00}\, dt^2 + 2g_{0i}\, dt\, dx^i + g_{ij}\, dx^i dx^j$, the radar distance between nearby stationary observers is the Landau–Lifshitz spatial metric*

$$\gamma_{ij} = g_{ij} - \frac{g_{0i} g_{0j}}{g_{00}}.$$

Proposition 14.14 (Synchronization holonomy; [25, 22]). *On a stationary slice, Einstein synchronization along a closed loop has holonomy $\oint g_{0i}/g_{00}\, dx^i$. For a ring of radius R rotating with angular velocity Ω ($c = 1$), the proper-time difference at the source is $4\Omega A/\sqrt{1 - \Omega^2 R^2}$ (Sagnac). It is zero iff the slice is static. Radar distance converges to γ_{ij} with error of order two in the separation (measured order 1.99); the inertial arc length differs by 15% at $\Omega R = 0.5$.*

No-Go 14.15 (Non-stationary slices). Without a timelike Killing field, radar distance depends on the emission time. For comoving observers in an FLRW slice with $a(t) = t^{2/3}$, it grows sixfold from $t = 1$ to $t = 16$ and differs from proper distance. No time-independent spatial metric is defined. Stationary slices are therefore the exact domain of the radar rungs.

Hypothesis 14.16 (H-med). The propagation speed in the protocol is c, or is anchored by a protocol whose own anchor is exact, or the slowness field is resolved by a ray family with trivial null space.

Proposition 14.17 (Media). *Delay data are line integrals of slowness. Without H-med, thickness and speed are exchangeable (the pair (d, v) has a dilation fibre), and an incomplete ray family leaves a nonzero invisible slowness sector [14].*

Hypothesis 14.18 (H-stat). Time translation belongs to Γ on the source channel across an anchor outage.

Theorem 14.19 (Ergodic transfer; [17]). *Under H-stat, the second and third moments of a stationary source's invariant measure determine the detector map across the outage. The ambiguity is the symmetry group of those moments; if that group consists of isometries, all distances are determined. Drift is tested by a clock-based period invariant that no spatial map changes. Unflagged drift bounds the residual error by a measured floor.*

Hypothesis 14.20 (H-val). The proportionality constant λ fixed by one anchor predicts a second, held-out anchor within a declared tolerance.

Theorem 14.21 (Physical distance). *Assume H-reg, H-nuis, H-SI, H-med, H-val, the Γ-conditions of Theorems 14.6 and 14.7, and translation membership of Proposition 14.12. Then the unique Γ-invariant metric h has distance d_{rad} in metres, and for every natural experiment $\mathcal{E}$ on the chart*

$$\bar{g}_{\mathcal{E}} = \lambda_{\mathcal{E}}^2 h, \qquad d_{\mathrm{phys}} = d_{RG}^{\mathcal{E}}/\lambda_{\mathcal{E}},$$

with $\lambda_{\mathcal{E}}$ constant. RG angles equal physical angles. In rank one the identification additionally requires the concatenation law of Imported Theorem 14.5.

Proof. Theorems 14.6 and 14.7 give $\bar{g}_{\mathcal{E}} = \lambda_{\mathcal{E}}^2 h$ with h determined by ρ. Radar count is a natural two-point scalar, hence a function of ρ (Theorem 14.6); by H-SI it is additive along collinear concatenations, hence linear (Imported Theorem 14.5). Proposition 14.11 fixes the constant in metres; Proposition 14.12 and H-med identify radar with rod distance; H-val excludes a response that fits one anchor only. In rank at least two Imported Theorem 14.9 makes one anchored distance sufficient. □

15 The assembled theorem

Theorem 15.1 (Kernel–stabilizer theorem). *Let $\mathcal{S}$ be an observation system with joint law F, kernel groupoid $\mathcal{K}$ and effective symmetry Γ, and let all modules considered be natural. On each regular stratum:*

1. *exact ignorance is $\mathcal{K}$; infinitesimal ignorance is $V = \operatorname{rad} g_F$;*
2. *every module descends to the quotient and is Γ-invariant;*
3. *under H-sound and H-coh, completeness is constant on Γ_G-orbits;*
4. *clocks with equal kernels form one transition groupoid; a time flow in* Stab *makes it affine;*
5. *$SO^+(1,n) \le \Gamma_q$, $n \ge 2$, forces the light cone and the conformal ray;*
6. *a counted channel excludes dilations from Γ; Euclidean diagonal gauge and relative Euclidean symmetry force one two-point invariant and $\bar{g}_{\mathcal{E}} = \lambda_{\mathcal{E}}^2 h$;*
7. *under the hypotheses of Theorem 14.21, h is physical distance in metres.*

Corollary 15.2 (Compatibility equations). *On every regular stratum claimed for a slice the following hold simultaneously:*

$$\operatorname{rad} g_F = V, \qquad \mathfrak{isom}(\bar{g}_{\mathcal{E}}) = \mathfrak{isom}(h), \qquad \Gamma_q C_q = C_q, \qquad \lambda_{\mathcal{E}} \text{ constant},$$

the clock transition groupoid is Γ-invariant, and the held-out anchor is predicted. Each equation depends on the same subgroups of Γ; if a subgroup is absent, every equation that uses it fails.

Corollary 15.3 (Stratified closure). *Under H-unif, the regular-stratum objects of Theorem 15.1 extend to the stratified quotient category of Definition 6.13:*

- *kernels are upper semicontinuous at walls;*

- *quotients specialize by maps under kernel inclusion, and by correspondences otherwise;*
- *symmetries descend through the normalizer of the wall kernel;*
- *distinguishability specializes under H-hell;*
- *characteristic sets are tame-typed, hyperbolically locked, or wild with window certificates;*
- *dynamical statements are typed certificates.*

Theorem 15.4 (Global transport)**.** *1. The observable atlas closes without extra data. Transport is the transition groupoid within strata and specialization across walls: maps under kernel inclusion, correspondences otherwise. Wall geometry comes from re-discrimination of the specialized task family.*

2. Latent transport is undetermined without declared data (Theorem 12.7). Under H-cov it is canonical, coherent across walls, and curved exactly by the latent holonomy (Theorem 12.9).

No internal open problem remains. What remains is external (tame type enumeration) or an extension (Section 21).

16 Certificates and closure

Definition 16.1 (Requested module and statuses)**.** A *request* is a declared domain, a requested module (for example a metric on a stratum, a cone type, a physical distance, or a transport), and its admissibility contract: the gates of the rungs it depends on. A request returns exactly one status:

- LOCKED: every gate is certified;
- ESCALATE: a gate is undecided and an admissible escalation remains;
- CORRESPONDENCE: several observationally indistinguishable objects remain, and the maximal correspondence is returned;
- VACUOUS: no nonzero admissible task survives;
- INCONSISTENT: a gate is contradicted;
- STRUCTURAL WALL[ε]: the gate remains undecided after the escalation budget, stated at the resolution ε reached;
- OUTSIDE DOMAIN: the request is not admitted to the declared domain.

The quantum wall outcomes map onto these statuses: DIVERGENT becomes STRUCTURAL WALL, since the task changes type, and UNDERDETERMINED becomes CORRESPONDENCE.

Theorem 16.2 (Totality and determinism)**.** *The dispatcher evaluates the gates in dependency order: domain, consistency, task, vacuity, rank, candidate, scale. On every combination of certificate outcomes it returns exactly one status, and LOCKED iff every gate passes.*

Proof. It is a finite decision procedure whose only LOCKED leaf is the conjunction of all gates. The battery checks all 384 combinations and reaches all seven statuses. □

Theorem 16.3 (Composite soundness)**.** *If gate g passes falsely with probability at most α_g, then* $\Pr(\textit{false LOCKED}) \le \sum_g \alpha_g$.

Proof. A false LOCKED requires at least one false gate pass; apply the union bound. □

The battery wires the dispatcher to the real certifiers: the double-escalation rank certificate (Proposition 8.5), the nonparametric Petz band (Theorem 7.21) and the declared anchor. On 5×400 instances with known truth it gave 0 false LOCKED and power 1.0 on the lockable instances. With the same dispatcher:

- numerical rank of noisy data gave 60–84 false LOCKED per seed;
- a panel argmin gave 16–32;
- a silently assigned scale gave 8–14.

Closure is exactly as sound as its weakest certificate [22].

Proposition 16.4 (Walls are window-indexed)**.** *A true wall and a full-rank law with $\sigma_{\min} < \tau$ both end as STRUCTURAL WALL[τ]. An exact wall is asserted only from the exact (model) layer.*

Proposition 16.5 (Thresholds form a monotone filtration; [67])**.** *For fixed data, the set of LOCKED requests can only shrink as stringency increases (smaller α, larger τ). In the test suite there were* 0 *violations over* 200 *instances and* 20 *threshold pairs. Each certificate can therefore be reported as its* critical margin*, the least stringent threshold at which it fails, which is a consistent statistic of the data. VACUOUS and OUTSIDE DOMAIN do not depend on thresholds. Thresholds themselves cannot be removed: at TV distance t, every decision rule has worst-case error at least $(1 - t)/2$ (Le Cam).*

Theorem 16.6 (Maximal closure over a declared domain)**.** *Fix a domain and a requested module with its admissibility contract. Every admitted request terminates in exactly one status of Definition 16.1. It is LOCKED only when every gate of the module is certified. Otherwise RG returns either the maximal observable correspondence or a typed abstention, and the false-lock probability is bounded by Theorem 16.3. In particular, no admitted request requires choosing between observationally indistinguishable alternatives.*

The theorem does not derive the domain, the task or the protocol, which remain declarations. Nor does it admit extensions outside the declared domain.

Proof. Totality and determinism (Theorem 16.2); soundness (Theorem 16.3); CORRESPONDENCE and the abstentions are exactly the non-LOCKED leaves. A choice between indistinguishable alternatives would be a LOCKED leaf with a failed candidate gate, which Theorem 16.2 excludes. □

17 Computational complexity

task	cost	source
kernel of a finite law table (n states, rows of length m)	$O(nm)$ hashing, union–find $O(n\,\alpha(n))$	exact equality oracle
joint kernel of k protocols	$O(knm)$	intersection (L3)
minimal sufficient panels	output can be $\binom{m}{\lfloor m/2 \rfloor}$; panel of size $\le k$: NP-complete	Shamir threshold [87]; Minimum Test Collection [86]
stratification of a semialgebraic law	doubly exponential in dimension	cylindrical decomposition [88, 89]
rank certificate in dimension d	$O(d^3)$ per escalation level	eigen-decomposition
Petz function at a ratio	$O(1)$ per response	Theorem 7.21
Čech gluing on a finite atlas	$O(\#\text{triples} \cdot d^3)$	cocycle check
dispatcher	$O(\#\text{gates})$	Theorem 16.2
BV forgetting lock, dyadic level L	$O(2^L n)$ exact rationals	Theorem 9.9
certified chaotic forecast, horizon n	$\ge n\lambda/\ln 2$ bits of initial data	Theorem 13.12
TV certificate without a class	impossible from finite samples	No-Go 9.11
Monte Carlo on a FORCED output	$O(\tau^{-2})$ draws (DKW)	[79]

Exactness is cheap at the exact layer and expensive in three specific places:

- *selection:* panel antichains are exponential, and minimum panels are NP-hard;
- *stratification:* doubly exponential in dimension;
- *chaos:* precision linear in horizon, at $\lambda/\ln 2$ bits per step.

Each cost sits where the theory reports it, not inside a declaration.

18 The ladder

Dependencies point only to earlier rungs. Status: F forced, I imported, H hypothesis-conditioned (named), Γ Γ-conditional, N no-go, O open.

Rung	Statement	Status	Depends	Ref.
Exact layer				
L0	Observation system, joint law, relabeling group	input	—	3.1
L1	Kernel groupoid; $Q \cong \operatorname{im} F$	F	L0	4.1
L2	Universal factorization; invariants are functions on Q	F	L1	4.2
L3	Joint kernel is the intersection; $Q \hookrightarrow \prod Q_i$ not onto	F	L1	4.3
L4	Garbling enlarges the kernel	F	L1	4.4
L5	Exact layer determines no higher structure	N	L2	4.5
Symmetry layer				
L6	$1 \to \text{Gau} \to \text{Stab} \to \Gamma \to 1$; liftable Γ faithful on Q (presentation-relative); $\Gamma \le \Gamma_{\text{obs}}$ (presentation-invariant)	F	L1	5.2
L7	The sequence need not split	N	L6	5.3
L8	Count rigidity	F	L0	5.7
L9	Orbit–quotient comparison	H: sound, compl	L1	5.9

Rung	Statement	Status	Depends	Ref.
L10	Rank, separation, completeness, reconstruction are distinct	N	L2, L9	5.10
L11	Coherent completeness	H: coh	L6, L9	5.11
Observable sector				
L12	Complete minimal-panel selector (finite index set); ties retained	F	L3, L11	5.12
L13	No task-free isotropic selector	N	L12	5.14
L14	More protocols intersect kernels; more models close their union; common decoder	F	L3, L12	5.15
L15	Coupled law refines its marginals; pairing is not coupling	F	L3	5.16
L16	Confidence-set bridge; point answers need a radius	F	L2, L14	5.17
L17	Tolerance is not a groupoid	N	L1	5.18
Differential layer				
L18	Tangent sequence; infinitesimal completeness defect	H: reg	L2, L9	6.2
L19	Natural descent: basic, Γ-invariant; radical $= V$ if nondegenerate on the image tangent	F	L6, L18	6.3
L20	Descent of law-defined structures is automatic	F	L19	6.4
L21	Singular strata: slices and tangent cones	I	L18	6.5
L22	First-order tangent misses one-sided directions at walls	F	L21	6.6
L23	Kernel upper semicontinuity; pointwise convergence and sampling insufficient	H: unif	L2, L22	6.8
L24	Specialization map iff kernel inclusion; correspondence otherwise; functorial	F	L4, L23	6.10
L25	Symmetry descent through the normalizer of $\mathcal{K}_0$	F	L6, L24	6.11
L26	Tangent upper semicontinuity; rank lower semicontinuity; finite tame strata	F / I	L18, L23	6.12
L27	Stratified quotient category; holonomy within strata	H: unif	L24, L26	6.14
Second-order layer				
L28	Fisher is natural; $\operatorname{rad} g_F = V$; Gaussian form	F	L19	7.2
L29	Naturality does not select the metric; Fisher under H-mono (Čencov)	N / H: mono	L19, L28	7.3
L30	Regular reconstruction of (R, N, Σ)	I	L28	7.6
L31	Fisher does not determine the covariance	N	L30	7.7
L32	Second-order quotient has dimension $p + q + pq$	F	L2, L30	7.8
L33	Second-order shadow is not the law	N	L32	7.9
L34	Efficient metric: conventions and nuisance wall	H: nuis	L28	7.12
L35	Data processing contracts Fisher	I	L4, L28	7.13
L36	Kernel order is not Blackwell order	N	L4, L35	7.14
L37	Distinguishability specializes under Hellinger convergence; weak is not enough	H: unif, hell	L27, L35	7.16
L38	Quantum metrics: Petz family gate; commuting collapse	I	L28, L35	7.18
L39	Quantum task lock: SLD, BKM, WY	H: task	L38	7.20
L40	Nonparametric identification of the Petz function; bands, not panels	F	L38	7.21
L41	Quantum wall specialization iff $f(0) > 0$; re-discrimination	F	L37, L39	7.22
L42	Trace rule (Gleason, Busch)	I	L38	7.23
L43	Contextual protocols: no hidden values; product law over contexts (Fine)	I / F	L15, L42	7.24

Rung	Statement	Status	Depends	Ref.
Ignorance				
L44	Persistent null; finite attainment	H: sens	L3, L28	8.2
L45	Sensor, precision and representation nulls	F	L44	8.3
L46	Finite-sample boundary reconstruction	I	L28, L34	8.4
L47	Rank walls are never certified from finite data	F	L26, L46	8.5
L48	Exact law equality is not uniformly certifiable from finite samples	N	L1, L47	8.6
Uncertainty, probability and forgetting				
L49	Uncertainty splits along the quotient; kernel prior never forgotten	F / I	L2, L28, L47	9.2
L50	Probability from exchangeability (de Finetti)	H: exch	L49	9.4
L51	Gaussian propagation is the LAN limit; branch selection at folds	I / F	L22, L49	9.5
L52	TV rigidity under bijections; fibre migration	N	L1, L49	9.6
L53	No uniform forgetting without a declared class	N	L52	9.7
L54	Mixing locks: compactness; sharp BV bound; logistic by conjugacy	H: adm	L53	9.9
L55	Contraction gate $G_N < 1$; Dobrushin blocks; functional type	H: adm	L53	9.10
L56	Samples refute, operators lock; accuracy gate	I / N	L53	9.11
L57	Non-circularity certificate; Monte Carlo placement	H: adm	L54–L56	9.12
L58	Non-dominated and infinite-dimensional strata (Feldman–Hájek)	I / F	L37, L49	9.13
Statistical distance				
L59	RG distance: representative- and chart-free, Γ-invariant, dimensionless	H: reg, nuis	L19, L34	10.2
L60	Garbling does not increase RG length	F	L35, L59	10.3
L61	No angle on rank-one strata	N	L59	10.4
L62	Finite divergences are not additive	N	L59	10.5
L63	RG distance is experiment-relative	N	L59, L60	10.6
Order and cone				
L64	Fibre-compatible reachability descends	F	L2	11.1
L65	Projective typing of the cone	N	L64	11.3
L66	Realization branches	H: top	L21, L65	11.4
L67	Quadratic cone determines a conformal ray	I	L66	11.5
L68	Lorentz isotropy forces the light cone ($n \geq 2$)	Γ	L6, L19, L66	11.6
L69	Tame reduction: finite types, sheets, integer dimension	I / F	L26, L66	11.7
L70	No finite linear classification	N	L69	11.8
L71	Smoothness and strict convexity are dual	I	L66	11.9
L72	Hyperbolic lock; residual symbol implies ill-posedness	I / F	L69, L71	11.10
L73	Wild class: resolution invariants, window certificates only	N	L69	11.11
Scale and transport				
L74	Scale representative iff exact cocycle; holonomy	H: int	L67	12.2
L75	Strict transport fixes the conformal factor	F	L74	12.3
L76	Induced connection on regular strata	I	L18	12.4
L77	Entropy does not select a connection	N	L2	12.5
L78	Law-defined transport descends; latent transport iff projectable	F / H	L20	12.6

Rung	Statement	Status	Depends	Ref.
L79	No law-natural connection: vertical shears act transitively	N	L19, L78	12.7
L80	Canonical connection under declared covariance; curvature, memory, wall coherence; physical under H-resp (Prop. 12.11)	H: cov; resp	L26, L79	12.9
Duration				
L81	Clock transition groupoid	F	L2	13.1
L82	Affine clocks under a stabilizer time flow	Γ	L6, L81	13.2
L83	Path independence as kernel inclusion; no orientation	F	L81	13.3
L84	Coherence duration and BCH memory	I	L81	13.4
L85	Finite-history noiseless experiments are stratified-regular	F	L27, L81	13.5
L86	Topological entropy from kernel counts	I	L2, L85	13.6
L87	Ruelle, Pesin and variational locks	I	L86	13.7
L88	Typed dynamical certificates; no zero-entropy certificate	F	L47, L87	13.8
L89	Cone fields of chaos: global only under uniform hyperbolicity	I	L72, L88	13.9
Chaos frontier				
L90	Frontier law: slope $1/\lambda$; protocols move intercepts	I / F	L47, L89	13.10
L91	Duality 2λ; quotient growth h	F	L86, L90	13.11
L92	Information cost $\lambda/\ln 2$ bits per certified step	I	L91	13.12
L93	Clock distortion classes: offset, rate, warp	F	L8, L81	13.13
Physical distance				
L94	No internal unit	N	L28, L59	14.1
L95	Counted channel removes dilation	F	L3, L6, L8	14.2
L96	Multiplicative count loss reopens the dilation fibre	N	L95	14.3
L97	Ratio scale	I	L6	14.5
L98	Two-point invariant	Γ	L6, L19	14.6
L99	Metric proportionality $\bar{g} = \lambda^2 h$	Γ	L19, L98	14.7
L100	Anchor budget $\dim(\mathrm{Sym}^2 V^*)^K$	Γ	L99	14.8
L101	Unit-distance rigidity ($n \geq 2$)	I	L99	14.9
L102	Radar distance; factor $\frac{1}{2}$	H: SI	L8, L81	14.11
L103	Radar equals rod distance iff no exchanged-frequency shift	Γ	L74, L95, L102	14.12
L104	Stationary slices: Landau–Lifshitz radar metric	I	L103	14.13
L105	Synchronization holonomy (Sagnac); zero iff static	F	L104	14.14
L106	Non-stationary slices: no time-independent radar metric	N	L104	14.15
L107	Media: dilation fibre and invisible slowness	H: med	L44, L102	14.17
L108	Ergodic transfer of the metre across outages	H: stat	L82, L99, L107	14.19
L109	Physical distance	H: reg, nuis, SI, med, val; Γ	L94–L107	14.21
Assembly				
L110	Kernel–stabilizer theorem; compatibility equations	Γ	L1–L109	15.1
Global transport and closure				
L111	Global transport: observable atlas closed; latent transport canonical under declared data	H: cov	L80, L110	15.4

Rung	Statement	Status	Depends	Ref.
L112	Statuses; totality and determinism of the dispatcher	F	L111	16.2
L113	Composite soundness: false LOCK $\leq \sum \alpha_g$	F	L47, L112	16.3
L114	Walls are window-indexed; thresholds form a monotone filtration (critical margins)	F	L47, L113	16.4
L115	Maximal closure over a declared domain	F	L112–L114	16.6
Open				
L116	Enumeration of tame characteristic types (Hilbert's sixteenth problem; external)	O	L69	21.2
L117	Uniform forgetting for forced deterministic turbulence above the gate	O	L55	21.5

19 Procedure

An application of RG to a physical slice consists of the following steps, each producing the object named.

1. Declare the protocols, the latent space and the relabeling group; form the joint law. (L0)
2. Determine the kernel by equality-of-law tests; obtain Q and its invariants. (L1–L3)
3. Test candidate generators of Stab by equality of laws after the declared operation and relabeling; record Γ. (L6)
4. Certify regular strata: constant rank of the law map, and of target and nuisance blocks; locate walls and use slices or jets there. (L18, L21, L34)
5. Compute natural modules: Fisher tensor, efficient metric, invariants, clock readouts, cone field. (L19, L28)
6. Read off the Γ-conditional structure: affine duration, cone type, two-point invariant, metric proportionality. (L68, L82, L98, L99)
7. Anchor: include a counted two-way timing channel; verify translation membership on the clock channel; resolve the medium. (L95, L102–L107)
8. At walls, specialize under declared uniform convergence (H-unif). Carry distinguishability only under Hellinger convergence. Report rank walls as upper bounds. (L23–L27, L37, L47)
9. For dynamics, use typed certificates on escalation plateaus, under the entropy locks. (L85–L89)
10. For quantum protocols, declare the task and identify the Petz function as a band; at support walls, re-discriminate. (L38–L41)
11. For latent transport, declare the second-order latent data or report the transport as undetermined. (L79, L80, L111)
12. For uncertainty, declare the admissible class and the functional type. Certify forgetting by an operator inequality and report kernel components as declared. (L49–L58)
13. Dispatch the request: report exactly one status of Definition 16.1, with its window.
14. Validate with a held-out anchor and check the compatibility equations. (L109, L110)

A failure at any step is attributed to one of: kernel misspecification; a wall; incomplete saturation; a smaller Γ; a non-natural module; a failed anchor, medium or scale hypothesis; statistical rejection; or latent transport used without declared second-order data (L111).

20 Status

Layer	Status
Exact, symmetry, natural descent	canonical relative to the specified system; law-space definitions selected by characterization (H-mono) or declaration
Strata and walls	forced under H-unif; distinguishability under H-hell; rank walls only bounded from data
Second order, Fisher, ignorance	forced or imported; nuisance under H-nuis
Quantum metric	family gate imported; member fixed by H-task; identified as a band from data
Cone and residual	realization branches; Lorentz under Γ; residual tame-typed, hyperbolically locked, or wild (window certificates)
Scale, transport	scale under H-int; connection not law-natural; canonical under H-cov
Duration, dynamics	clock groupoid forced; dynamical statements as typed certificates
Physical distance	under H-reg, H-nuis, H-SI, H-med, H-val and Γ; ergodic transfer under H-stat
Global	kernel–stabilizer theorem, stratified closure, global transport
Closure	every admitted request ends in exactly one status; LOCKED only with all gates certified; false-lock rate bounded by the gate errors
Observable sector	minimal-panel antichain, robust quotients, common decoders, confidence-set bridge
Uncertainty	primitive is indistinguishability; probability by exchangeability, mixing on a declared class, or quantum structure; non-circularity certificate
Chaos	frontier slope $1/\lambda$, duality 2λ, quotient growth h, cost $\lambda/\ln 2$ bits per step
Extensions	contextual quantum (Fine), non-dominated, infinite-dimensional Gaussian (Feldman–Hájek), non-stationary (no-go): closed
Open	tame type enumeration (external); forced deterministic turbulent forgetting

21 Open problems

Remark 21.1 (Closed: latent connection). The former rung asking for a canonical latent connection is closed by Theorems 12.7 and 12.9. No connection is law-natural, and under declared second-order latent data the connection is canonical.

Open Problem 21.2 (Tame type enumeration)**.** The residual branch of Theorem 11.4(5) is reduced by Section 11.1 to tame types (finite data), the hyperbolic lock, and the wild class (no finite classification). What remains is to enumerate the tame types of degree-d cones. For plane curves this is the first part of Hilbert's sixteenth problem, known for $d \leq 7$ [45]; it is a problem of real algebraic geometry, not of RG.

Remark 21.3 (Closed: extensions). The former extension problem is closed:

- contextual quantum protocols by Theorem 7.24;
- non-dominated and infinite-dimensional Gaussian experiments by Proposition 9.13;
- non-stationary slices by No-Go 14.15.

General non-Gaussian infinite-dimensional experiments follow the same pattern through Kakutani's dichotomy for product measures, which is not developed here.

Remark 21.4 (Programme: crossing into physics). Physical values are certified as values conditional on a computed residual set of assumptions, by agreement of routes with disjoint remaining assumptions. Contradicting clusters return CORRESPONDENCE and are resolved only by registered, powered tests of their distinctive assumptions. Physical-curvature effects are typed by resolution (DETECTED or STRUCTURAL WALL[τ]). The stress tensor is an RG target: LOCKED where a geometric route and a source route agree, an explicit correspondence otherwise, and the field equation is the consilience test between the two. The quantum–geometry passage is certified by the final task image on the complete admissible fibre (source, coupling, renormalization, boundary and calibration alternatives). Identifying every intermediate link is sufficient but not necessary. Within declared models, mean geometry from raw second-moment panels is LOCKED (conditional), backreaction closes under contraction, and a finite-shot witness excludes the separable null synthetically. Coupling selection and full curved-background semiclassical closure remain open. See *Real Measurements*, *Crossing into Physics*, *The Stress Tensor* and *The Quantum Bridge* [68]. This is a research programme, not a rung.

Open Problem 21.5 (Forced deterministic turbulence)**.** For forced deterministic Navier–Stokes above the gate $G_N \geq 1$, no rung certifies uniform forgetting. TV is rigid (No-Go 9.6), the contraction gate fails, and there is no noise and no expanding non-invertible transfer operator. With stochastic forcing, ergodicity is known qualitatively [82]. An explicit uniform constant over a declared class is open.

Remark 21.6 (Computational witnesses). Each No-Go and each finite example in this paper has a deterministic numerical witness in the accompanying scripts, including the specialization, residual, chaotic, quantum, transport, closure, wall–tap, groupoid, chaos-frontier, uncertainty, forgetting, observable-sector and final-lock batteries [19, 18, 17, 20, 21, 22, 54, 55, 56, 57, 58, 59, 62]. Each battery carries mutation canaries: a check that cannot fail is not counted. The witnesses check the stated counterexamples; they are not premises of any proof.

References

[1] S. Amari, H. Nagaoka, *Methods of Information Geometry*, AMS/Oxford, 2000.

[2] F. S. Beckman, D. A. Quarles, On isometries of Euclidean spaces, *Proc. AMS* 4 (1953), 810–815.

[3] D. Blackwell, Comparison of experiments, *Proc. 2nd Berkeley Symp.*, 1951.

[4] E. Buckingham, On physically similar systems, *Phys. Rev.* 4 (1914), 345–376.

[5] N. N. Čencov, *Statistical Decision Rules and Optimal Inference*, AMS, 1982.

[6] G. E. Crooks, Measuring thermodynamic length, *Phys. Rev. Lett.* 99 (2007), 100602.

[7] M. Goresky, R. MacPherson, *Stratified Morse Theory*, Springer, 1988.

[8] S. W. Hawking, A. R. King, P. J. McCarthy, A new topology for curved space–time which incorporates the causal, differential, and conformal structures, *J. Math. Phys.* 17 (1976), 174–181.

[9] O. Hölder, Die Axiome der Quantität und die Lehre vom Mass, *Ber. Sächs. Ges. Wiss.* 53 (1901), 1–64.

[10] D. H. Krantz, R. D. Luce, P. Suppes, A. Tversky, *Foundations of Measurement, Vol. I*, Academic Press, 1971.

[11] L. Le Cam, *Comparison of Experiments*, IMS, 1996.

[12] D. Luna, Slices étales, *Bull. Soc. Math. France, Mémoire* 33 (1973), 81–105.

[13] D. B. Malament, The class of continuous timelike curves determines the topology of spacetime, *J. Math. Phys.* 18 (1977), 1399–1404.

[14] L. Pestov, G. Uhlmann, Two dimensional compact simple Riemannian manifolds are boundary distance rigid, *Ann. Math.* 161 (2005), 1093–1110.

[15] M. J. Pflaum, H. Posthuma, X. Tang, Geometry of orbit spaces of proper Lie groupoids, *J. reine angew. Math.* 694 (2014), 49–84.

[16] M. Rada, *Operational Duration*, 2026.

[17] M. Rada, *Resolution Geometry on Chaotic and Dirty Experiments*, 2026.

[18] M. Rada, *Residual Characteristic Geometry*, 2026.

[19] M. Rada, *Stratified Specialization*, 2026.

[20] M. Rada, *Quantum Metric Selection*, 2026.

[21] M. Rada, *Canonical Transport*, 2026.

[22] M. Rada, *Maximal Closure*, 2026.

[23] L. D. Landau, E. M. Lifshitz, *The Classical Theory of Fields*, 4th ed., Pergamon, 1975, §84.

[24] V. Perlick, On the radar method in general-relativistic spacetimes, in *Lasers, Clocks and Drag-Free Control*, Springer, 2008, 131–152.

[25] E. J. Post, Sagnac effect, *Rev. Mod. Phys.* 39 (1967), 475–493.

[26] D. Petz, Monotone metrics on matrix spaces, *Linear Algebra Appl.* 244 (1996), 81–96.

[27] S. L. Braunstein, C. M. Caves, Statistical distance and the geometry of quantum states, *Phys. Rev. Lett.* 72 (1994), 3439–3443.

[28] F. Hiai, D. Petz, The proper formula for relative entropy and its asymptotics in quantum probability, *Comm. Math. Phys.* 143 (1991), 99–114.

[29] T. Ogawa, H. Nagaoka, Strong converse and Stein's lemma in quantum hypothesis testing, *IEEE Trans. Inf. Theory* 46 (2000), 2428–2433.

[30] K. M. R. Audenaert et al., Discriminating states: the quantum Chernoff bound, *Phys. Rev. Lett.* 98 (2007), 160501.

[31] J. Calsamiglia et al., Quantum Chernoff bound as a measure of distinguishability between density matrices, *Phys. Rev. A* 77 (2008), 032311.

[32] R. E. Kalman, A new approach to linear filtering and prediction problems, *J. Basic Eng.* 82 (1960), 35–45.

[33] H. E. Rauch, F. Tung, C. T. Striebel, Maximum likelihood estimates of linear dynamic systems, *AIAA J.* 3 (1965), 1445–1450.

[34] B. O'Neill, The fundamental equations of a submersion, *Michigan Math. J.* 13 (1966), 459–469.

[35] L. Gårding, Linear hyperbolic partial differential equations with constant coefficients, *Acta Math.* 85 (1951), 1–62.

[36] L. Gårding, An inequality for hyperbolic polynomials, *J. Math. Mech.* 8 (1959), 957–965.

[37] R. Hardt, Semi-algebraic local-triviality in semi-algebraic mappings, *Amer. J. Math.* 102 (1980), 291–302.

[38] L. Le Cam, G. L. Yang, *Asymptotics in Statistics*, 2nd ed., Springer, 2000.

[39] F. Ledrappier, Some properties of absolutely continuous invariant measures on an interval, *Ergodic Theory Dynam. Systems* 1 (1981), 77–93.

[40] M. Misiurewicz, W. Szlenk, Entropy of piecewise monotone mappings, *Studia Math.* 67 (1980), 45–63.

[41] J. Renegar, Hyperbolic programs, and their derivative relaxations, *Found. Comput. Math.* 6 (2006), 59–79.

[42] R. T. Rockafellar, *Convex Analysis*, Princeton, 1970.

[43] D. Ruelle, An inequality for the entropy of differentiable maps, *Bol. Soc. Brasil. Mat.* 9 (1978), 83–87.

[44] L. van den Dries, *Tame Topology and o-minimal Structures*, Cambridge, 1998.

[45] O. Viro, Real plane algebraic curves: constructions with controlled topology, *Leningrad Math. J.* 1 (1990), 1059–1134.

[46] M. Wojtkowski, Invariant families of cones and Lyapunov exponents, *Ergodic Theory Dynam. Systems* 5 (1985), 145–161.

[47] M. Rada, *The Law and Boundary of Observational Ignorance*, 2026.

[48] M. Rada, *Observation Laws, Universal Quotients, and Saturated Observable Geometry*, 2026.

[49] M. Rada, *Resolution Geometry: Reconstruction and Identifiability*, 2026.

[50] H. Reichenbach, *Philosophie der Raum-Zeit-Lehre*, de Gruyter, 1928.

[51] BIPM, *The International System of Units (SI)*, 9th ed., 2019.

[52] F. Zhang (ed.), *The Schur Complement and Its Applications*, Springer, 2005.

[53] N. T. Zung, Proper groupoids and momentum maps: linearization, affinity, and convexity, *Ann. Sci. Éc. Norm. Supér.* 39 (2006), 841–869.

[54] M. Rada, *Resolution Geometry of the Wall–Tap Experiment*, 2026.

[55] M. Rada, *The Groupoid of Resolution*, 2026.

[56] M. Rada, *Resolution Geometry Does Not Hide Chaos: The Frontier Law*, 2026.

[57] M. Rada, *Uncertainty Geometry*, 2026.

[58] M. Rada, *Uncertainty Geometry: The Forgetting Ladder*, 2026.

[59] M. Rada, *The Observable Sector from Joint Laws*, 2026.

[60] M. Rada, *Resolution Geometry: Source-Side Automation*, 2026.

[61] M. Rada, *From the Wall to the Kernel*, 2026.

[62] M. Rada, *Final Locks* (test suite FL01–FL08), 2026.

[63] M. Rada, *Physical Curvature: The Canonical Connection as a Law of Motion* (test suite PC01–PC11), 2026.

[64] A. Shapere, F. Wilczek, Geometry of self-propulsion at low Reynolds number, *J. Fluid Mech.* 198 (1989), 557–585.

[65] R. Montgomery, Gauge theory of the falling cat, *Fields Inst. Commun.* 1 (1993), 193–218.

[66] M. Rada, *Conditional Rigidity* (test suites UM01–UM06, UX01–UX09), 2026.

[67] M. Rada, *Presentation and Thresholds* (test suite DF01–DF07), 2026.

[68] M. Rada, *Real Measurements* (RD01–RD10), *Crossing into Physics* (CN01–CN11) *The Stress Tensor* (ST01–ST11) and *The Quantum Bridge* (QB01–QB10), 2026.

[69] B. de Finetti, La prévision: ses lois logiques, ses sources subjectives, *Ann. Inst. H. Poincaré* 7 (1937), 1–68.

[70] A. M. Gleason, Measures on the closed subspaces of a Hilbert space, *J. Math. Mech.* 6 (1957), 885–893.

[71] P. Busch, Quantum states and generalized observables: a simple proof of Gleason's theorem, *Phys. Rev. Lett.* 91 (2003), 120403.

[72] A. Fine, Hidden variables, joint probability, and the Bell inequalities, *Phys. Rev. Lett.* 48 (1982), 291–295.

[73] N. D. Mermin, Simple unified form for the major no-hidden-variables theorems, *Phys. Rev. Lett.* 65 (1990), 3373–3376.

[74] A. W. van der Vaart, *Asymptotic Statistics*, Cambridge, 1998.

[75] A. Lasota, J. A. Yorke, On the existence of invariant measures for piecewise monotonic transformations, *Trans. AMS* 186 (1973), 481–488.

[76] A. Lasota, M. C. Mackey, *Chaos, Fractals, and Noise*, Springer, 1994.

[77] R. L. Dobrushin, Central limit theorem for nonstationary Markov chains, *Theory Probab. Appl.* 1 (1956), 65–80.

[78] H. Weyl, Über die Gleichverteilung von Zahlen mod. Eins, *Math. Ann.* 77 (1916), 313–352.

[79] P. Massart, The tight constant in the Dvoretzky–Kiefer–Wolfowitz inequality, *Ann. Probab.* 18 (1990), 1269–1283.

[80] J. Feldman, Equivalence and perpendicularity of Gaussian processes, *Pacific J. Math.* 8 (1958), 699–708.

[81] J. Hájek, On a property of normal distributions of any stochastic process, *Czech. Math. J.* 8 (1958), 610–618.

[82] M. Hairer, J. C. Mattingly, Ergodicity of the 2D Navier–Stokes equations with degenerate stochastic forcing, *Ann. Math.* 164 (2006), 993–1032.

[83] V. I. Oseledets, A multiplicative ergodic theorem, *Trans. Moscow Math. Soc.* 19 (1968), 197–231.

[84] A. A. Brudno, Entropy and the complexity of the trajectories of a dynamical system, *Trans. Moscow Math. Soc.* 2 (1983), 127–151.

[85] Ya. B. Pesin, Characteristic Lyapunov exponents and smooth ergodic theory, *Russian Math. Surveys* 32 (1977), 55–114.

[86] M. R. Garey, D. S. Johnson, *Computers and Intractability*, Freeman, 1979 (problem SP6).

[87] A. Shamir, How to share a secret, *Comm. ACM* 22 (1979), 612–613.

[88] G. E. Collins, Quantifier elimination for real closed fields by cylindrical algebraic decomposition, LNCS 33, Springer, 1975, 134–183.

[89] J. H. Davenport, J. Heintz, Real quantifier elimination is doubly exponential, *J. Symbolic Comput.* 5 (1988), 29–35.